



Design and Evaluation of an Interactive Visual Analytics System for Explainable Artificial Intelligence

Master's Thesis

by

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Declaration of Authorship

I hereby declare that the submitted master's thesis titled "**Design and Evaluation of an Interactive Visual Analytics System for Explainable Artificial Intelligence**" is my own work.

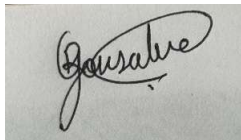
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A handwritten signature in black ink, appearing to read 'Gourav', is written on a light-colored rectangular piece of paper.

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Abstract

Though machine learning models are being adopted more and more to aid in business decision-making, their predictions are frequently hard for non-technical users to grasp. This can diminish trust and complicate the practical use of model results. This thesis showcases the creation and assessment of an interactive visual analytics system aimed at explainable artificial intelligence. It integrates a customer churn prediction model, explanations based on SHAP, and an interactive dashboard in Power BI.

The IBM Telco Customer Churn dataset was used. This data was cleaned then transformed and then prepared for machine learning. Several classification models were trained and compared. I selected the gradient boosting tuned model as it gave a proper stability between prediction of the performance and identifying of customers who were at chance of leaving. The final decision threshold was adjusted from 0.50 to 0.27 to improve churn recall. Using this threshold, the model achieved an accuracy of 73.06%, a churn recall of 80.75%, an F1-score of 0.6144, and a ROC-AUC score of 0.8383.

Here the SHAP was used to explain the model at a global and an individual customer level. The generated prediction and explanation results were prepared as structured files and imported into Power BI. A six-page dashboard was developed to present customer risk, overall churn drivers, individual explanations, customer segments, and model performance.

The system was evaluated by comparing the interactive dashboard with a static explanation baseline. The interactive version produced higher task accuracy and stronger ratings for understanding, usefulness, trust, and decision support. The findings show that combining machine learning explanations with an interactive visual interface can make complex model results easier to explore and interpret. The main contribution of this thesis is the integration of prediction, explainability, visual analytics, and user evaluation in one practical workflow.

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List of Abbreviations

Abbreviation	Meaning
AI	Artificial Intelligence
BI	Business Intelligence
CRISP-DM	Cross-Industry Standard Process for Data Mining
CSV	Comma-Separated Values
DAX	Data Analysis Expressions
EDA	Exploratory Data Analysis
F1-score	Harmonic Mean of Precision and Recall
FN	False Negative
FP	False Positive
KPI	Key Performance Indicator
ML	Machine Learning
PR-AUC	Area Under the Precision–Recall Curve
ROC-AUC	Area Under the Receiver Operating Characteristic Curve
SHAP	SHapley Additive exPlanations
TN	True Negative
TP	True Positive
UI	User Interface
XAI	Explainable Artificial Intelligence

Chapter 1: Introduction

1.1 Background

As we know that Machine learning is becoming an important part of future and present data analysis. Most of the Organisations use the predictive models to study the historical data, recognise patterns and support decisions. All these models are used in the areas such as customer management, finance, healthcare, marketing, and risk analysis. In most cases the result of the models is presented as a predicted class, score, or probability. So basically, the model doesn't directly tell that why was that result produced. This is mainly known as a black-box problem [1].

Most of the predictions can be correct and on point but still be difficult for a person to understand. This always becomes a practical problem when a business user should explain a decision or decide whether the model result should be accepted or not. For example, a model may identify a customer as likely to leave a company, but the user may also need to know and have an idea that which customer characteristics contributed to that prediction. Because without this information, it can be difficult to decide that exactly what action should be taken.

Explainable Artificial Intelligence that is commonly known as XAI, it aims to make machine learning models and their predictions much easier to understand. XAI methods can provide information about the factors that influence a model. Some explanations describe the overall behaviour of the model, while others explain one individual prediction. One widely used method is SHAP, which assigns a contribution value to each feature involved in a prediction [2]. These values can help users understand which factors increased or decreased the predicted outcome.

Providing an explanation does not automatically mean that users will make better decisions. The way the explanation is presented is also important. Previous research has shown that confidence information and local explanations can influence how people trust an AI system, but explanations alone may not always improve decision quality [3]. This means that explanations should be understandable, relevant, and presented in a form that supports the user's task.

Visual analytics provides a useful way to present such information. These visuals combine the automated data analysis with the interactive visual interfaces thus allowing the users to explore results, compare values, apply filters, and investigate important patterns [4]. Instead of only receiving a prediction or a technical explanation the users can interact with the generated information and thus examine it from different viewpoints.

This thesis applies these ideas to customer churn prediction. A machine-learning model is used to estimate whether a telecommunications customer is likely to leave the company. So here SHAP is used so that the results of the models can be explained properly. The predictions and explanations are presented through an interactive Power BI dashboard. The purpose is not only to create an accurate prediction model, but also to make its results easier for non-technical users to understand and use in decision making.

1.2 Motivation

Customer churn is an important business problem in industries where customers can easily change from one service provider to another. When a customer leaves, the company may lose future income and may also need to spend additional resources to attract a replacement. Because of these reasons the churn prediction models are used to identify the customers who have a high chance of leaving [5]. This information can help a company focus its retention activities on customers who may require attention.

However, the prediction alone may not provide enough information for a useful business decision. A model may classify a customer as high risk, but the employee using the result may still want to know why the customer received that classification. Old and recent researches on churn modelling did show that predictive performance should be considered together with the comprehensibility and practical usefulness of the model [5]. An exactly higher result may be difficult to apply when the reasoning behind it is not exactly clear.

This issue becomes more important when complex machine-learning models are used. These models may identify relationships that are difficult to explain using normal business rules. Recent churn research has therefore started to combine prediction models with explainability methods. Such studies use global explanations to identify the main factors affecting the model and local explanations to show why one specific customer received a particular prediction [6]. These explanations can make the result more meaningful, but they may still be difficult for users to understand when presented only through technical plots or programming outputs.

Another motivation for this thesis was the need to connect model explanations with an interface that business users can explore. Data scientists often examine predictions and SHAP values inside Python notebooks. This format is useful during model development, but it is not always suitable for employees who mainly work with reports and dashboards. Visual analytics offers an opportunity to present the same information through charts, filters, customer selections, and other interactive elements [4].

Power BI was selected because it is widely used for business reporting and allows several related views to be connected in one dashboard. In this thesis, it is used to present churn predictions, risk levels, model performance, global feature effects, individual customer explanations, and customer segment comparisons. The dashboard is designed manually so that the analytical results produced in Python can be shown in a form that is easier to examine.

The final motivation was to evaluate whether interaction actually improves the use of model explanations. Creating an attractive dashboard does not prove that users understand the model better. Therefore, this thesis compares the interactive Power BI system with a static explanation baseline. This comparison mostly focuses on the task accuracy, time of the completion, understanding and trusting, usefulness and supporting decisions. This makes all of it possible to evaluate not only the machine-learning model, but also the way its predictions and explanations are communicated to users.

1.3 Problem Statement

The Machine learning models support customer retention decisions by identifying the customers who are going to or will leave a service provider. But need to keep in mind that a prediction score alone does not explain the reason behind the result. This creates a gap between the technical output of the model and the information required by a business user.

The problem is not limited to model accuracy. A model may produce useful predictions while still being difficult to understand. When the reason behind a prediction is hazy then the users may either ignore the result or trust it without proper examination. So, both situations can cause or lead to bad decision. Previous research has therefore treated model comprehensibility as an important part of churn prediction, especially when predictions are used to support business actions [5].

Explainability methods can reduce this problem by showing how different features contributed to a prediction. SHAP, for example, can provide both an overall view of the model and a detailed explanation for one customer. However, SHAP outputs are usually created inside programming environments and may contain technical charts, transformed feature names, and numerical values that are not immediately clear to non-technical users [2], [6].

A further problem concerns the presentation of these explanations. Most of the Static charts shows important outputs, but they provide lesser support for exploring different customers, customer groups, and risk factors. An interactive dashboard allows users to filter the data, compare groups, select an individual customer, and

examine different explanation views. Nevertheless, the presence of interaction does not automatically guarantee better understanding or better decisions. Research on

AI assisted decision-making has shown that explanations can influence trust, but their effect depends on how the information is presented and used [3].

Therefore, the central problem addressed in this thesis is the lack of an integrated and evaluated workflow that connects customer-churn prediction, model explanation, and interactive business visualisation in a form suitable for non-technical users. The research must address three related requirements. First, the system should produce reliable churn predictions. Second, it should explain the factors influencing the model at both global and individual levels. Third, these results should be presented in an interactive interface and evaluated against a static alternative.

The thesis creates a comprehensive workflow to tackle this issue, utilizing machine learning, SHAP, and Power BI. The system is then evaluated to determine whether the interactive dashboard helps users understand the predictions, identify important factors, and make decisions more effectively than a static explanation.

1.4 Research Aim and Objectives

The main aim of this thesis is to design, implement, and evaluate an interactive visual analytics system that helps users understand machine learning predictions. The system combines a customer churn prediction model, SHAP explanations, and a Power BI dashboard.

The study focuses on the practical use of explainable artificial intelligence in a business setting. It does not only examine how accurately the model predicts customer churn. It also studies how the prediction results and explanations can be presented in a form that is easier for users to understand.

To achieve this aim, the thesis follows several objectives.

1. To examine and prepare the Telco Customer Churn dataset for machine learning.
2. To train and compare different classification models for customer churn prediction.
3. To select and tune a final model based on its performance and its ability to identify customers who are at risk of leaving.
4. To apply SHAP to explain the overall behaviour of the model and individual customer predictions.

5. To prepare the model predictions and explanation results in a structured format for use in Power BI.
6. To design an interactive dashboard that presents customer risk, important churn factors, individual explanations, customer groups, and model performance.
7. To compare the interactive dashboard with a static explanation baseline.
8. Need to evaluate the system by using task accuracy and completion time and by understanding, trust, usefulness, and decision support.

Together all of these objectives covered the complete process from data preparation to model development, explanation, dashboard creation, and user evaluation. This will enable my project to examine both the model's technical performance and the practical benefit of displaying all results through an interactive visual interface.

1.5 Research Questions

The research questions were developed from the main aim of the thesis. These all questions focus on model performance, explainability, dashboard design, and user evaluation.

RQ1: How effectively can the machine learning models predict the customer churn using the Telco Customer Churn dataset?

This question examines the performance of different classification models. All these models are compared using accuracy, precision, recall, F1 score, ROC AUC, PR AUC, and the confusion matrix. Then the final model is selected based on both general performance and its ability to find out customers who have chances of leaving.

RQ2: Which customer characteristics have the greatest effect on churn predictions, both across the overall model and for individual customers?

This question is addressed using SHAP. Global explanations are used to identify the most important features across the complete dataset. Local explanations are used to show how individual customer characteristics increase or decrease the predicted probability of churn.

RQ3: How can we present the Shap outputs and the machine learning predictions by showing an interactive Power BI dashboard in an understandable format for non technical users?

This question focuses on the design of the Power Bi analytics system. It examines how model results, customer risk, important features, individual explanations, customer groups, and model performance can be organised across different dashboard pages.

RQ4: Does an interactive visual analytics dashboard support better understanding and decision making than a static explanation?

This question is examined through the user evaluation. The interactive dashboard is compared with a static baseline using task accuracy, completion time, understanding, trust, usefulness, and decision support.

These questions cover the complete direction of the thesis. The first two questions focus on the technical results of the model and SHAP analysis. The third question focuses on dashboard design. The fourth question focuses on the practical value of the interactive system.

1.6 Scope of the Thesis

This thesis focuses on the design and evaluation of an explainable machine learning system for customer churn analysis. The work covers the complete process from data preparation to model development, explanation, dashboard creation, and user evaluation.

The study uses the Telco Customer Churn dataset as the main data source. The dataset contains customer information related to services, contracts, payment methods, and account history. The purpose of the analysis is to predict whether a customer is likely to leave the company.

Several classification models are trained and compared. The final model is selected based on its predictive performance and its ability to identify customers who are at risk of churn. The study also considers the effect of changing the decision threshold because identifying more churn customers is important for retention activities.

SHAP is used to explain the model results. The thesis includes global explanations that describe the overall behaviour of the model and local explanations that describe individual customer predictions. The explanation results are prepared in a structured form and connected to Power BI.

The dashboard is limited to the six pages developed during the project. These pages present the overall business view, customer risk, global model explanations, individual customer explanations, customer segment comparison, and model performance. The dashboard is intended for users who understand business analytics but may not have advanced knowledge of machine learning or programming.

The evaluation compares the interactive dashboard with a static explanation. The study measures task accuracy, completion time, understanding, trust, usefulness, and decision support. The purpose is to examine whether the interactive version helps users work with model explanations more effectively.

The thesis does not include the deployment of the model in a live company environment. It also does not include real time data processing, automatic retention actions, or direct connection to a company customer management system. The results are based on one public dataset and one business case. So basically, the findings should not be treated as a proof that the same system will give same results in every industry or organisation.

These borders keep the study focused on the practical connection between machine learning, explainability, visual analytics, and user evaluation.

1.7 Proposed Methodology

The research followed an applied development and evaluation process. The work started with understanding the customer churn problem and identifying how machine learning explanations could be presented to users. It then moved through data analysis, model development, explainability, dashboard design, and user evaluation.

The general structure was guided by the main stages of CRISP DM. This process includes business understanding, data understanding, data preparation, modelling, evaluation, and deployment [7]. The stages are connected, which means that results from a later stage can lead to changes in an earlier stage.

The first stage focused on understanding the business problem. Customer churn was selected because it is a suitable classification problem and has a clear business use. The main objective was to identify customers who may leave and explain the reasons behind the prediction.

The second stage examined the Telco Customer Churn dataset. To grasp the variables, customer segments, missing values, and churn distribution, exploratory data analysis was employed. Cleaning and preparation were done of the data for machine learning. Numerical and categorical variables were transformed into a proper suitable format and then the data was divided into training and testing sets.

The modelling stage compared several classification algorithms. Then the performance was checked using the accuracy, precision, recall, F1 score, ROC AUC, PR AUC and confusion matrix values. The final model chosen was the tuned Gradient

Boosting model. The prediction threshold was also adjusted to improve the identification of customers who were likely to churn.

After selecting the model, SHAP was used to study how different features influenced its predictions. Global explanations were created to show the general behaviour of the model. Individual explanations were also created to show why a particular customer received a specific churn probability.

The prediction results and SHAP values were prepared as structured data files. These files were imported into Power BI and used to create six dashboard pages. The dashboard included an executive overview, customer risk analysis, global explanations, individual explanations, customer segment comparison, and model performance information.

The final stage evaluated the dashboard. Participants completed similar tasks using a static explanation and the interactive Power BI system. Their answers, completion times, ratings, and preferences were collected and compared. This evaluation was used to examine whether interaction improved understanding, usefulness, trust, and decision support.

This methodology connects the technical development of the model with the practical presentation and evaluation of its results. It allows the thesis to examine the complete path from raw customer data to an interactive explanation system.

1.8 Main Contributions

The main contribution of this thesis is the development of a complete workflow that connects machine learning, explainable artificial intelligence, and interactive visual analytics. The work does not treat prediction, explanation, and visualisation as separate tasks. Instead, they are combined in one system that begins with customer data and ends with an interactive dashboard for business users.

The first contribution is the development and comparison of several customer churn classification models. The final Gradient Boosting model was selected after evaluating different performance measures. The decision threshold was also changed from 0.50 to 0.27 so that the model could identify a larger number of customers who were likely to leave. This decision was based on the importance of recall in customer retention activities.

The second contribution is the use of SHAP to explain the final model. The analysis includes both overall model behaviour and individual customer predictions. The explanation process was designed to make the results easier to understand. Processed

model features were connected back to the original customer variables, which reduced 45 transformed features into 19 business friendly feature groups.

The third contribution is the preparation of prediction and explanation outputs in a structured form that could be used directly in Power BI. This created a clear connection between the Python based machine learning process and the business intelligence environment.

The fourth contribution is the design of a six page Power BI dashboard. The dashboard presents an executive overview, customer risk analysis, global model explanations, individual customer explanations, customer segment comparison, and model performance. Each page was created for a different analytical purpose while remaining connected through filters and customer selections.

The final contribution is the evaluation of the interactive dashboard against a static explanation baseline. The comparison measures task accuracy, completion time, understanding, trust, usefulness, and decision support. This evaluation makes it possible to study whether the interactive system provides practical value beyond the model results alone.

Together, these contributions show how an explainable machine learning model can be transformed into a visual system that is more suitable for users with limited technical knowledge.

1.9 Structure of the Thesis

This thesis is divided into six different chapters.

Chapter one introduces the research background, motivation, problem statement, aim, objectives, research questions, scope, methodology and main contributions.

Chapter two gives the background and related research about machine learning, customer churn prediction, explainable artificial intelligence, SHAP, visual analytics, Power BI and user evaluation.

Chapter three explains the dataset and methodology covering data preparation, model development, threshold selection, SHAP analysis, dashboard design and the user evaluation method.

Chapter four presents the results, from machine learning models threshold analysis, SHAP explanations the Power BI dashboard and user evaluation.

Chapter five discusses the meaning of the findings answers the research questions and presents the limitations of the study.

Chapter six summarises the findings, contributions, practical implications, future work and final conclusion. The thesis ends with the reference list.

Chapter 2: State of the Art

2.1 Artificial Intelligence and Machine Learning

Artificial intelligence is a new and vast field of computer science it studies that how the computer systems can perform tasks that normally require normal to high human knowledge. These tasks can include reasoning, learning, understanding language, recognising patterns, and making decisions. The field contains several methods and research areas, including machine learning, natural language processing, computer vision, robotics, and knowledge based systems [8].

Machine learning is one of the main areas within the broader field of artificial intelligence. It allows a computer system that how to learn patterns from data instead of depending only on rules written by a programmer. Helps for being on its own. The system uses previous examples to build a model and then applies the old learned relationships to new data. This approach is useful when the relationship between the input and the expected result is too unique or complex to describe using a small number of fixed rules [9].

A machine learning project normally includes several connected stages. Data is collected and examined before it is prepared for modelling. A suitable algorithm is then trained using part of the available data. The trained model is tested using observations that were not used during training. This helps to determine whether the model has learned useful patterns or has only remembered the training examples [9].

Machine learning approaches are grouped into supervised, unsupervised, and reinforcement learning. The supervised learning trains the model using data that contains both input variables and a known result. The purpose is to learn the relationship between these inputs and the result. Unsupervised learning does not use a known target and instead searches for structures or groups within the data. Reinforcement learning involves an agent that learns through actions and feedback from an environment [8].

This thesis uses supervised learning because the Telco Customer Churn dataset contains a known outcome for every customer. The outcome shows whether the customer remained with the company or left. The customer info is taken as the input and then it learns to predict this outcome for new observations.

Machine learning does decision making by processing many variables and identifying relationships that are difficult to find through manual analysis. The model is just not useful simply because it produces a prediction. Its performance depends on the data quality and evaluation process, and the way the result is interpreted. Also, different algorithms may learn different patterns from the same dataset [9].

More complex models can sometimes achieve better predictive performance, but they may also be more difficult to understand. A user may receive a predicted class or probability without knowing which information had the strongest effect on that result. This creates an important connection between machine learning and explainable artificial intelligence. The need to understand model behaviour is especially relevant when predictions are used to support decisions involving customers, risks, or business actions.

In this thesis, machine learning provides the predictive part of the system. Explainability methods are then used to examine the model, while Power BI is used to communicate the results through an interactive visual interface. The following sections discuss supervised classification and customer churn prediction in more detail.

2.2 Supervised Classification

Supervised learning uses labelled examples to learn a relationship between input variables and a known target. Each observation contains a set of features together with the correct result. During training, the algorithm studies these examples and adjusts its internal parameters. The completed model can then produce a result for new observations that were not included in the training data [10].

Supervised learning problems are normally divided into regression and classification. Regression is used when the target is a numerical value, such as sales revenue or house price. Classification is used when the target belongs to a category. Examples include whether a transaction is fraudulent, whether an email is unwanted, or whether a customer will leave a company [10].

Customer churn prediction is a binary classification problem because the target contains two possible classes. First class shows the customers who remain with the company and the other class shows customers who leave. In this dataset that was used for the thesis these outcomes are represented by the values No and Yes.

A classification model may provide both a predicted class and a probability. The probability shows that the estimated likelihood that an observation belongs to a specific class. A decision threshold is later used to convert the probability into a final class. For example, if there is a threshold of 0.50 a customer with a churn probability above this value is taken as churn.

The selection of the threshold can affect the number of right and wrong predictions. A higher threshold makes the model more careful when predicting the positive class. Lowering the threshold helps identify more positive cases, but it can also lead to more

customers being incorrectly labelled as positive. So, in conclusion the threshold should be selected according to the purpose of the analysis instead of accepting a fixed value in every situation.

Classification algorithms learn relationships in different ways. Logistic Regression estimates the probability of a class through a mathematical relationship between the target and the input variables. Decision Trees divide the data using a series of conditions. Ensemble models combine several trees or learning stages to create a stronger prediction model. These differences mean that the same dataset can produce different results depending on the selected algorithm [11].

The model should be evaluated using data that was not included during training. This matters because good results on the training data do not always mean that the model will perform equally well for new customers. A model that learns the training examples too closely may fail to generalise. So basically, this is known as overfitting [11].

Classification performance should not be judged using accuracy alone. This is especially important when the two target classes are not equally represented. In a churn dataset, customers who stay may greatly outnumber those who leave. A model may look accurate because it predicts the larger class properly while failing to identify many churn customers. Because of this several evaluation measures are used together in this thesis.

Supervised classification provides the main technical foundation for the customer churn model. However, the model must also be understandable and useful for the intended users. The following section discusses customer churn prediction and how machine learning has been applied to this business problem.

2.3 Customer Churn Prediction

Customer churn is seen when a customer stops using the products or services of a company. In the telecommunications sector, this may happen when a customer cancels a contract or changes to another provider. Churn can affect company revenue and long term customer relationships. For this reason, companies try to identify customers who may leave before the cancellation takes place [12].

Customer churn prediction uses historical customer data to estimate the likelihood that a customer will leave. The available information may include contract type, length of service, monthly charges, payment method, internet service, technical support, and other service related details. A machine learning model studies the relationship between these variables and the known churn outcome.

The main purpose of churn prediction is to support customer retention activities. A company can use the prediction results to identify customers who may require attention. However, the model does not decide which action should be taken. The final decision mostly depends on customer value and available resources and also business rules, and the reason behind the predicted risk.

Customer churn can be predicted using methods such as Logistic Regression, Decision Trees, Random Forest, Gradient Boosting, Support Vector Machines, and neural networks. Their results vary because each method learns patterns in a different way. All results are different because each method identifies patterns differently. The results are also affected by data preparation, feature selection, parameter settings, class distribution, and the chosen evaluation method [5], [12].

A common challenge is that churn datasets usually contain fewer customers who leave than those who stay. So, a model shows high accuracy even when it fails to detect many customers who actually goes. To judge its performance properly the measures such as recall, precision, F1 score, ROC AUC, PR AUC, and the confusion matrix should be considered together.

The most useful measure depends on the purpose of the prediction. In customer retention, failing to detect a customer who actually leaves can be costly. This error is called a false negative. Reducing the decision threshold may help identify more churn customers, but it can also produce more false positives. Therefore, the threshold should be selected by considering both missed churn cases and unnecessary retention efforts.

Predictive performance is not the only concern in churn modelling. Business users may also need to understand why a customer has been classified as high risk. Verbeke and colleagues noted that the comprehensibility of churn models is important because the results must support practical decisions rather than provide only an accuracy value [5]. More recent work has combined churn prediction with explanation methods that provide both global and individual views of model behaviour [6].

The study by Poudel, Pokharel, and Timilsina used explainable machine learning to examine customer churn in telecommunications. Their work provided global explanations for the overall model and local explanations for individual predictions. This is relevant to the present thesis because it shows that churn prediction can be supported by information about the factors influencing each result [6].

However, an explanation produced in a programming environment may still be difficult for a business user to explore. The present thesis extends the practical workflow by

preparing the prediction and SHAP results for an interactive Power BI dashboard. This allows users to examine churn risk, important factors, individual customers, customer groups, and model performance through connected visual pages.

2.4 Explainable Artificial Intelligence

Explainable Artificial Intelligence is an area of research that aims to make the behaviour and results of artificial intelligence systems easier for people to understand. It is commonly referred to as XAI. The need for explainability has increased because many machine learning models can produce useful predictions without clearly showing how the result was reached [1].

Some models are naturally easier to understand than others. For example, a small Decision Tree can be followed through a sequence of conditions. A simple Logistic Regression model can also provide information about the relationship between its input variables and the predicted result. More complex models may contain many connected calculations, trees, or processing stages. Their internal behaviour can therefore be difficult to examine directly.

A model that hides its internal reasoning from the user is often described as a black box model. This does not mean that the model contains no logic. It means that the relationship between the inputs and the output is too complex to be easily understood by a person. This lack of transparency may become a problem when a prediction affects an important decision or requires an explanation [1].

XAI methods aim to address this issue by showing which factors affected a prediction. An explanation can indicate the most important features, whether they raised or lowered the outcome, and how the model responded to a specific case. This information can help users examine the model rather than accepting the prediction without question.

The purpose of XAI is not limited to increasing user trust. Explanations can also support model development and validation. A data scientist may use them to identify unexpected relationships, possible data problems, bias, or model behaviour that does not agree with domain knowledge. An explanation may therefore help both the person developing the model and the person using its results.

The Explainable Artificial Intelligence programme developed by the defence Advanced Research Projects Agency encouraged the creation of AI systems whose behaviour could be understood and used responsibly by human users [13]. This programme helped increase research interest in explanations that are useful to human users rather than only technically correct.

Explainability can be studied at different levels. Some methods examine the complete model and describe its general behaviour. Other methods focus on one prediction and explain why the model produced that particular result. Explanations may also depend on the internal structure of a specific algorithm, or they may work with different models by examining their inputs and outputs. These differences have produced a wide range of explanation methods [1], [14].

An explanation should not be treated as a complete description of everything happening inside a model. Many explanation methods provide an approximate view of the relationships that influenced a prediction. The quality of an explanation can also depend on the selected method, data, model, and user. For this reason, explanation results should be examined carefully and presented with sufficient context.

Explainability is important for the customer churn system developed in this thesis. A churn probability can show the estimated risk that a customer will leave, but it does not show which customer characteristics influenced that risk. SHAP is used in this study to provide this additional information. The explanations are then presented in Power BI so that users can explore the model at both the overall and individual customer levels.

2.5 Global and Individual Explanations

Machine learning explanations can be grouped into global and individual explanations. The main difference is whether the aim is to understand the model as a whole or a single prediction [1].

A global explanation presents the overall behaviour of a model. It can identify the features with the greatest general influence and show how changes in those features relate to the predicted result. This type of explanation is useful for understanding the main patterns learned by the model.

In customer churn analysis, a global explanation may identify contract type, customer tenure, monthly charges, or support services as important factors. However, it cannot fully explain why a specific customer received a high or low churn probability.

An individual explanation examines one observation. It shows how the feature values of that case contributed to the prediction. This helps users understand why a particular customer was classified as likely to churn.

For example, a short contract may increase the predicted churn risk for one customer, while a long relationship with the company may lower it. The final result depends on the combined effect of several customer characteristics.

Methods such as LIME explain individual predictions by building a simpler model around the selected observation [15]. SHAP also provides feature contributions for individual cases and can combine these values across many observations to describe the model's overall behaviour [2].

Global and individual explanations answer different analytical questions. A global view shows what the model has learned from the full dataset, while an individual view explains the result for one customer. For this reason, both are included in the system developed in this thesis.

The Power BI dashboard displays global feature importance on one page and individual customer explanations on another. This arrangement allows users to first understand the general model behaviour and then examine the reasons behind a selected customer prediction.

Table 1 Comparison of global and individual explanations

Aspect	Global explanation	Individual explanation
Main purpose	Understand the complete model	Understand one prediction
Main question	What generally influences churn?	Why did this customer receive this result?
Data level	Complete dataset or many observations	One selected observation
Example output	Overall feature importance	Feature contribution for one customer

2.6 SHAP

SHAP stands for SHapley Additive exPlanations. It is a method used to explain how individual input features influence the output of a machine learning model. SHAP was introduced by Lundberg and Lee as a common framework for explaining different types of predictive models [2].

The method is based on Shapley values from cooperative game theory. In the original game theory setting, Shapley values are used to distribute the result of a group fairly among its members. In machine learning, the features are treated as the members of the group and the model prediction is treated as the result. The contribution of each feature is estimated by examining how the prediction changes when different combinations of features are considered [2].

A SHAP explanation begins with a base value. This value represents the average model output for the reference data. Each feature then receives a SHAP value that shows how it moves the prediction away from the base value. A positive value increases the model output, whereas a negative value lowers it. This relationship is shown in the following equation.

$$\textit{Model output} = \textit{Base value} + \textit{Sum of feature contributions}$$

The size of a SHAP value shows the strength of the contribution. A larger absolute value means that the feature had a stronger effect on the prediction. The direction of the value shows whether the feature increased or decreased the model output.

For example, a month to month contract may increase the predicted churn probability for one customer. A long tenure may reduce the probability. The final churn prediction is produced by combining the effects of all customer features with the base value.

SHAP can be used for both individual and overall model analysis. At the individual level, the values explain one prediction. At the overall level, the absolute SHAP values from many customers can be combined to identify the features that generally have the strongest influence on the model. Research on tree models has shown that combining individual SHAP explanations can provide a broader understanding of global model behaviour [16].

One advantage of SHAP is that it provides a consistent form of feature contribution across different observations. This makes it possible to compare why different customers received different predictions. However, SHAP values still require careful interpretation. They describe how the model uses the available features, but they do not prove that a feature directly causes customer churn.

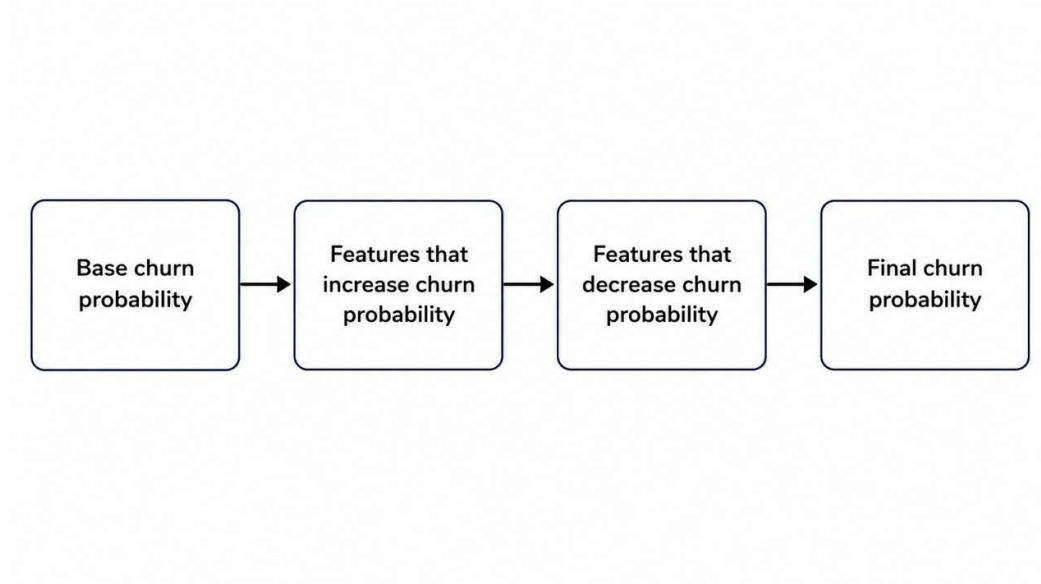
In this thesis, SHAP was applied to the selected Gradient Boosting model. The explanations were produced on the probability scale so that the contributions could be understood as changes in churn probability. This was easier to present to users than technical model outputs.

The preprocessing process created 45 model features. Some original variables produced several transformed features during encoding. These related values were grouped back into 19 original customer features. This made the explanations easier to understand and avoided showing several separate values for the same business concept.

The SHAP results were prepared for use in Power BI. Global results were used to show the most important churn factors across the dataset. Individual results were used to explain the prediction for a selected customer. This allowed the same explanation method to support both general model understanding and detailed customer analysis.

Figure 1 Basic structure of an individual SHAP explanation

Source: Author generated based on Lundberg and Lee [2].



2.7 Visual Analytics

Visual analytics brings analytical methods and interactive visual displays together. Computer-based techniques process the data, while users examine the results using charts, filters, selections, and linked views. Its main aim is to support reasoning and decision-making when the amount or complexity of the data makes manual analysis difficult [4].

A standard data visualisation usually communicates information through a chart or graph. Visual analytics offers more because users can actively explore what is shown. They can adjust filters, select observations, compare groups, and move between general summaries and detailed information. The analysis therefore develops through repeated interaction between the user and the system.

The field draws knowledge from areas such as information visualisation, statistics, data mining, data management, and human-computer interaction [4]. These areas perform different but connected functions. Statistical and machine-learning techniques discover patterns, visual displays make the findings easier to see, and interaction allows users to investigate them according to their analytical needs.

The user remains a central part of this process. Computers can calculate results and analyse many records quickly, but they may not understand the business meaning behind every finding. A person can use experience, context, and subject knowledge when interpreting the results. Visual analytics therefore joins automated processing with human judgement.

Visual analytics also has an important role in machine learning. Earlier research has shown that interactive visual methods can assist several stages of the machine-learning process. These include exploring the data before modelling, examining the model during development, and understanding its results after training [17].

Within explainable artificial intelligence, visual analytics can help users investigate why a model produced a certain result. A user may first examine a general view of the most influential features and then explore one individual observation. Filters and selections can also be used to compare model behaviour across different customer groups.

We should keep in mind that the design of a visual analytics system should reflect the requirements of its intended users. Technical users may need detailed model information whereas business users may require more clearer explanations and easy summaries. Presenting too much information on one page can make the system confusing. The design must therefore maintain a suitable balance between detail, readability, and interaction.

In this thesis, Power BI serves as the visual analytics environment. The dashboard connects churn predictions, probabilities, SHAP explanations, customer details, and model performance measures. Users can begin with the complete customer population, examine specific groups, and then study an individual customer.

Model training is not carried out within Power BI. The machine-learning model and SHAP calculations are completed in Python. Their output files are then imported into Power BI and displayed through interactive dashboard pages. This approach allows Python to perform the analytical work, while Power BI supports exploration and communication.

Visual analytics therefore links the technical model with the people who use its results. It converts prediction and explanation values into information that users can explore and apply when making customer-retention decisions.

2.8 Interactive Dashboards and Power BI

A dashboard brings important information together through a set of connected visual elements. It has controls and data displays including charts, tables, indicators, filters, and links. The objective of an interactive dashboard is to combine relevant data into a single dashboard view enabling the user to grasp a situation quickly or research areas of importance. This gives users more control over the analysis than a static image or printed report.

Dashboard design is not only about placing several charts on a page. The information should be arranged according to the questions users need to answer. Research on dashboard design suggests that interaction can create an analytical conversation between users and data. Users should be able to examine a result, change their focus, and continue the analysis without losing the meaning of the information [18].

It is one of the Business Analytics Services which is from the house of Microsoft, that's Power BI. It enables the ability to connect to sources, perform calculations and display data on interactive reports [19]. The reports are populated by graphics such as cards, charts, tables, matrices, slicers which can be inter linked for one to update the display on others.

Slicers are filtering controls placed directly on a report page. They allow users to filter the data without opening a separate filter panel. This makes common selections more visible and easier to use [20]. In this thesis, slicers are used to select individual customers and explore different customer groups.

Microsoft distinguishes between a Power BI report and a Power BI dashboard. A report may contain several interactive pages, while a Power BI dashboard is normally a single page created in the Power BI service [21]. The system developed in this thesis is technically a six page Power BI report created in Power BI Desktop. However, the term dashboard is used throughout the thesis in a broader business sense because the full report works as one connected visual analytics system.

The system includes six pages. The Executive Overview page shows the main churn figures and business indicators. The Customer Risk Analysis page helps users examine customers across different risk levels. The Global Model Explanation page presents the factors with the strongest overall influence on the model.

The Individual Customer Explanation page allows users to select a customer and examine the factors that raised or lowered the predicted churn probability. The Customer Segment Comparison page supports comparisons between customer groups. The Model Performance Summary page presents evaluation measures and the results of the selected decision threshold.

All dashboard pages are connected through a shared data model. Filters and customer selections help keep the information consistent across related visual elements. The layout was designed manually in Power BI. Python was used to generate predictions and SHAP values, while Power BI was used to organise and present the results.

The interactive design also has some limitations. Users may become confused if a page includes too many charts or controls. Filters can also change displayed values without the user noticing the active selection. Clear titles, consistent page layouts, visible filters, and simple navigation are therefore important design features.

In this thesis, the dashboard is not used only to present final results. It forms part of the explainability system. It allows users to move from an overall model view to a detailed explanation for one selected customer. The user evaluation later compares this interactive system with a static explanation to examine whether interaction improves understanding and decision making.

Table 2 Comparison of static and interactive presentation

Aspect	Static presentation	Interactive Power BI report
User control	Information is fixed	Users can select and filter data
Level of detail	One prepared view	Summary and detailed views
Customer analysis	Limited to displayed examples	Different customers can be selected
Group comparison	Fixed comparison	Customer groups can be filtered
Navigation	No page interaction	Connected report pages
Main purpose	Communicate prepared findings	Support exploration and analysis

2.9 Understanding, Trust, and Decision Support

Even though machine learning models can provide accurate predictions, they can be problematic to interpret by the end-user. Here, interpretation is how well a user can interpret what is being predicted, how well a user can interpret what data is being displayed, and how well the user can interpret why the prediction was made a certain way. In a perfectly explainable system, the user must be able to figure out what the model predicted and why this particular feature had an impact on the prediction.

Understanding is important because users may need to explain a result or decide what action should be taken. In customer churn analysis, a user may need to know whether a customer has a high churn risk and which customer characteristics contributed to that risk. A probability alone may not provide enough information for this decision.

Trust describes the willingness of a user to rely on the information produced by a system. Trust should not simply be as high as possible. A user should rely on the model when it provides useful and reliable information, but should also question the result when the available evidence is unclear. Research on trust in automated systems describes this as appropriate reliance [22].

Too little trust may cause users to ignore useful predictions. Too much trust may cause them to accept a result without checking whether it is reasonable. The goal of an explainable system is therefore to support a level of trust that matches the actual quality and limits of the model.

Explanations can influence how users view a model, but they do not always improve decision accuracy. Zhang, Liao, and Bellamy found that confidence information could help users adjust their trust in an artificial intelligence model. However, improved trust adjustment did not automatically lead to better combined decisions between the user and the model [3].

This finding is important because an explanation should not be evaluated only by asking whether users liked it or trusted it. It is also necessary to examine whether users understood the information and completed the required tasks correctly. This is reflected in the evaluation for the thesis by having user ratings along with the task results that are to be measured. Decision support is a question of supplying information in such a manner as to aid a human to come to a judgment. The user is not removed from the task, and no decision is made by the system as to how to act and retain.

Instead, it presents churn risk, model explanations, customer details, and group information that can support the final decision.

The interactive dashboard may support decision making by allowing users to move between summary and detailed information. A user can first examine the overall churn situation, then identify important factors, and finally investigate one customer. This process allows the user to combine model information with personal judgement and business knowledge.

The static explanation used in the evaluation presents prepared information without interaction. The Power BI system allows users to select customers, apply filters, compare groups, and move between connected pages. The evaluation examines whether this additional control improves task accuracy, understanding, trust, usefulness, and decision support.

Understanding, trust, and decision support are related, but they are not the same. A user may understand a prediction without trusting it. A user may also trust a system without fully understanding how it works. Measuring these areas separately provides a clearer view of how the dashboard supports its users.

Table 3 Main concepts used in the user evaluation

Concept	Meaning in this thesis	How it is examined
Understanding	Ability to interpret the prediction and explanation	Task answers and participant ratings
Trust	Willingness to rely on the system appropriately	Participant rating
Decision support	Help provided for making a customer related judgement	Task performance and participant rating

2.10 Related Work

Previous research related to this thesis can be divided into three main areas. The first area focuses on customer churn prediction. The second area adds explainability methods to churn models. The third area uses interactive visual analytics to help users understand machine learning systems.

Verbeke and colleagues examined the importance of producing churn models that are both accurate and understandable. Their study compared rule based methods for churn prediction and showed that model comprehensibility is important when prediction results must be used for customer retention decisions [5]. This work is relevant because it treats practical understanding as an important part of model quality rather than focusing only on predictive accuracy.

More recent research has included explanation methods within customer churn analysis. Poudel, Pokharel, and Timilsina compared several machine learning models for telecommunications churn and reported that Gradient Boosting produced the strongest result in their experiment. They used SHAP and other visual methods to provide overall and individual explanations of the model [6]. Their study showed that explainability can help identify important churn factors and describe the reason behind a particular customer prediction.

The present thesis has a similar interest in combining churn prediction with SHAP. However, it also focuses on how the explanation results are prepared and presented to users. The SHAP outputs are transformed into structured data and displayed through a six page Power BI report. The system also includes a formal comparison between a static explanation and an interactive version.

Interactive explainability has also been studied through visual analytics systems. Spinner and colleagues proposed explAIner, a framework that combines different explanation methods with interactive visual views. The system was designed to support model understanding, model comparison, the identification of model limitations, and trust building. It was evaluated with nine participants who had different levels of experience [23].

The explAIner study is relevant because it shows that explainability can be treated as an interactive process. Users are not limited to viewing one prepared explanation. They can examine different parts of the model and continue their investigation according to the information they find. However, explAIner was developed within TensorBoard and mainly supported the machine learning development process. It did not focus on customer churn or Power BI.

Zöller and colleagues developed XAutoML, an interactive visual analytics system for understanding machine learning pipelines created through automated machine learning. The system combined explanation methods with interactive visualisations and was connected to JupyterLab. Its evaluation included users from different professional and technical backgrounds. The study reported that participants were able to gain useful information and improve their understanding of the generated machine learning pipelines [24].

XAutoML demonstrates the value of combining explanations with interactive visual analysis. However, its main purpose was to explain the process and output of automated machine learning. The intended users also included data scientists and machine learning researchers. The present thesis has a different focus because it presents churn predictions and explanations to users who may have more experience in business analytics than in model development.

The reviewed studies show that customer churn prediction, SHAP explanations, and interactive visual analytics have already been investigated separately and in some combined forms. Therefore, this thesis does not claim that these individual concepts are new. Its contribution lies in the specific design and evaluation of a complete workflow that connects a tuned churn model, probability based SHAP explanations, grouped business features, a six page Power BI report, and a comparison with a static explanation.

Within the studies reviewed for this thesis, no study presented the same complete combination of customer churn prediction, SHAP explanation data, Power BI presentation, individual customer analysis, segment comparison, and a user evaluation against a static baseline.

2.11 Comparison of Existing Studies

The studies discussed in the previous section address different parts of the research problem. Some focus mainly on customer churn prediction, while others examine model explanations, interactive visual analysis, or user trust. Comparing these studies helps to show which parts are already established and where the present thesis differs.

Verbeke and colleagues focused on the balance between predictive performance and model comprehensibility in customer churn prediction. Their work showed that understandable churn models are useful when predictions need to support customer retention decisions. However, the study did not use SHAP or an interactive dashboard [5].

Poudel, Pokharel, and Timilsina combined telecommunications churn prediction with local and global explanations. Their study compared several classification models and reported Gradient Boosting as the strongest model in their experiment. SHAP plots and other visual methods were used to explain important churn factors. The study is closely related to the modelling and explanation parts of this thesis, but it did not present the results through a Power BI system or compare static and interactive explanations [6].

Spinner and colleagues developed explAiner as an interactive visual analytics framework for explainable machine learning. The system supported model understanding, comparison, diagnosis, and refinement through several connected explanation views. A user study with nine participants was conducted to examine the proposed workflow. The system was developed within TensorBoard and was not designed specifically for customer churn or business reporting [23].

Zöller and colleagues introduced XAutoML to make automated machine learning processes and generated pipelines more transparent. The system combined explainability methods with interactive visualisations and was connected to JupyterLab. Its evaluation showed that users could obtain useful information and improve their understanding of automated machine learning pipelines. However, the system focused on technical model development rather than customer related business decisions [24].

Zhang, Liao, and Bellamy examined the effect of confidence information and local explanations on trust in artificial intelligence assisted decisions. Their results showed that explanations and confidence information can influence trust, but better trust adjustment does not always produce better decision performance [3]. This finding supports the decision to measure both participant ratings and task accuracy in the present thesis.

The comparison shows that previous research provides useful foundations for churn prediction, model explanation, visual analytics, and user evaluation. However, these elements are usually studied separately or within systems aimed at technical users. The present thesis brings them together in one applied workflow for customer churn analysis.

Table 4 Comparison of related studies

Study	Main focus	Prediction or explanation method	Interactive system	User evaluation	Difference from this thesis
Verbeke et al. [5]	Comprehensible churn prediction	Rule based churn models	No	No	No SHAP or Power BI system
Poudel et al. [6]	Explainable telecom churn prediction	Gradient Boosting and SHAP	Limited to explanation plots	No dashboard comparison	No Power BI report or static comparison
Spinner et al. [23]	Interactive explainable machine learning	Several XAI methods	explAIner in TensorBoard	Nine participants	Not focused on churn or business users
Zöller et al. [24]	Understanding automated machine learning	XAI methods for AutoML pipelines	XAutoML in JupyterLab	User study	Focused on technical AutoML analysis
Zhang et al. [3]	Trust in artificial intelligence decisions	Confidence and local explanations	Experimental interface	Yes	No churn model or Power BI dashboard
Present thesis	Churn prediction and interactive explanation	Gradient Boosting and SHAP	Six page Power BI report	Static and interactive comparison	Combines prediction, explanation, dashboard, and user evaluation

The comparison indicates that the closest study to the modelling part of this thesis is the work of Poudel and colleagues because it combines telecommunications churn prediction with SHAP explanations. The closest studies to the interactive part are explainer and XAutoML. However, neither system uses Power BI or focuses on presenting individual customer churn explanations to business users.

Within the literature reviewed for this thesis, no study was found that combined the same model development process, probability based SHAP outputs, grouped customer features, six connected Power BI pages, and a comparison between static and interactive explanations. This does not mean that the individual methods are new. The difference lies in how they are connected and evaluated within one system.

2.12 Research Gap

The reviewed literature shows that customer churn prediction, explainable artificial intelligence, and visual analytics are well established research areas. Several studies have developed accurate churn models, while others have used SHAP and similar methods to explain model behaviour. Interactive systems have also been created to help users examine machine learning models [6], [23], [24].

However, the reviewed studies do not fully address the practical connection between model development, explanation, business visualisation, and user evaluation. Churn studies often focus on model performance and explanation plots. These outputs are useful for data scientists, but they may not be easy for business users to explore.

Interactive explainability systems such as explainer and XAutoML provide several connected views and support detailed investigation [23], [24]. However, these systems are mainly designed for model developers, researchers, or users working in technical environments such as TensorBoard and JupyterLab. They do not focus on customer churn analysis inside a commonly used business intelligence platform.

Another gap concerns the presentation of individual explanations. A global importance chart can show which features are generally important, but it cannot fully explain one customer prediction. Some churn studies provide local SHAP plots, but these are usually shown as separate technical visualisations rather than as part of a connected dashboard that also includes customer information, risk level, segment analysis, and model performance [6].

There is also a gap in the evaluation of interactive explanation systems for business users. A dashboard may look informative, but this does not prove that it improves understanding or decision making. The effect of interaction should be examined using both user opinions and measurable task results. Previous research has shown that explanation and trust do not always lead to better decision performance [3].

This thesis addresses these gaps by developing a complete workflow for customer churn analysis. The workflow begins with data preparation and machine learning model development. It then applies SHAP at the global and individual levels. The results are prepared in a structured format and presented through a six page Power BI report.

The system brings together customer risk analysis, global explanations, individual explanations, customer segment comparison, and model performance. It is designed for users who may understand business analytics but may not have advanced programming knowledge.

The final part of the research compares the interactive Power BI report with a static explanation. The evaluation measures task accuracy, completion time, understanding, trust, usefulness, and decision support. This allows the study to examine whether interaction provides practical value beyond the presentation of model explanations alone.

The contribution of this thesis therefore lies in the integration and evaluation of existing methods rather than the invention of a new machine learning or explanation algorithm. Based on the literature reviewed in this chapter, the same complete combination of customer churn prediction, probability based SHAP explanations, grouped business features, Power BI presentation, and static versus interactive evaluation was not identified. This statement is limited to the studies reviewed and does not claim that no similar unpublished system exists.

2.13 Chapter Summary

This chapter presented the theoretical background and previous research related to the thesis. It began by explaining artificial intelligence, machine learning, and supervised classification. These concepts provide the technical foundation for the customer churn prediction model developed in this study.

Customer churn prediction was then discussed as a business classification problem. Previous research showed that model performance is important, but the results must also be understandable and useful for customer retention decisions.

The chapter introduced Explainable Artificial Intelligence as a way to make model behaviour and individual predictions easier to examine. Global explanations describe the general behaviour of a model, while individual explanations focus on one selected prediction.

SHAP was presented as the main explanation method used in this thesis. SHAP values show how each feature moves a prediction away from the average model output. The method can support both overall feature analysis and individual customer explanations.

Visual analytics and interactive dashboards were also discussed. These methods combine automated analysis with visual exploration. Power BI was selected as the presentation environment because it supports connected pages, filters, slicers, charts, tables, and customer selections.

The chapter also examined understanding, trust, and decision support. Previous research showed that explanations may influence user trust, but they do not automatically improve decision quality. This supports the use of both task performance and participant ratings in the evaluation.

The related work comparison showed that previous studies have examined churn prediction, SHAP, interactive explainability, and user trust. However, the reviewed studies did not present the same complete combination of probability based SHAP explanations, grouped customer features, a six page Power BI report, and a comparison between static and interactive explanations.

Chapter 3: Dataset and Methodology

3.1 Research Design

This thesis used an applied research design. The purpose was to develop a working visual analytics system and evaluate whether it helped users understand machine learning predictions. The research therefore included both technical development and user evaluation.

The technical part of the study began with customer data. The data was examined, cleaned, and prepared before different classification models were trained. The models were compared using several performance measures. A tuned Gradient Boosting model was selected and used to produce customer churn probabilities and predicted classes.

The selected model was then explained using SHAP. The explanation process examined the overall behaviour of the model and the prediction made for each customer. The SHAP results were connected with the original customer information and prepared as structured files for Power BI.

Power BI was used to create the visual part of the system. Six connected pages were designed to present customer risk, important churn factors, individual customer explanations, customer segments, and model performance. The report allowed users to filter information, select customers, and move between general and detailed views.

The research process was guided by the main stages of CRISP DM. These stages include business understanding, data understanding, data preparation, modelling, evaluation, and deployment [7]. The stages were not treated as a strict one way process. Results from later stages were used to review and improve earlier decisions.

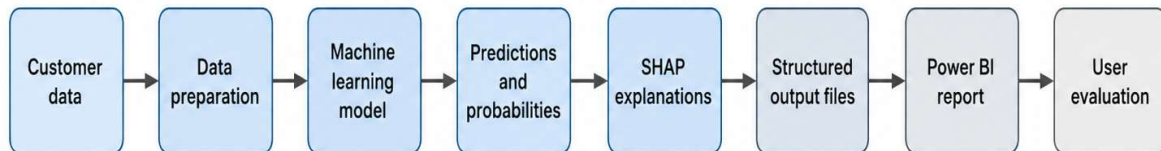
For example, the model evaluation influenced the selection of the final decision threshold. The SHAP results also influenced how the model features were grouped and prepared for Power BI. Dashboard development required further changes to the structure of the exported data so that the visual elements could respond correctly to user selections.

The evaluation part compared two ways of presenting the same analytical information. The first version was a static explanation that did not allow interaction. The second version was the interactive Power BI report. Participants completed similar tasks with both versions.

The comparison used task accuracy, completion time, understanding, trust, usefulness, and decision support. Participant preferences and comments were also reviewed. This made it possible to examine whether the interactive system provided practical benefits beyond the machine learning model and SHAP results.

The complete research design connected customer data, prediction, explanation, visual presentation, and user evaluation in one workflow. This structure followed the direction defined during the research planning stage and was maintained throughout the completed project.

Figure 2 Complete research workflow used in the thesis



3.2 Dataset Description

The dataset used in this thesis is the Telco Customer Churn dataset. It was originally provided as an IBM sample dataset and was accessed through Kaggle. The data represents a fictional telecommunications company that provides telephone and internet services to its customers [25], [26]. IBM describes the dataset as a sample for studying whether customers left the company during the previous month.

The dataset comprises 7,043 records of customers and 21 column features of the dataset. Each record of the dataset is of an individual customer, whereas every feature explains an aspect of the relationship between the customer and the company. Variables the features range from, but are not limited to: customer account number, demographics of customer, service subscription, account management information, payments and the classification to either yes/no depending on whether a customer chose to stop using the service and left the company in an event known as churn or if [26].

The last variable (last column, in our case is the outcome) is Churn which may have only one value from the pool. Either it means customer decided to stop with our services or if else this client decide to stay with our company.

In our case study, the number of customers who were the churn (Yes) are 1,869 whereas 5,174 of customer were none. So, we have an estimated amount of 26.5 percent of customer that will be classified as churn. The remaining customers belong to the non churn class.

The difference between the two class sizes is important for model evaluation. A model could predict the larger non churn class for most customers and still obtain a reasonable accuracy. However, such a model would fail to identify many customers who were actually at risk of leaving. For this reason, the study does not depend only on accuracy. Values which were also taken into view were Recall, precision, F1 score, ROC AUC, PR AUC, and confusion matrix.

The demographic variables include gender, senior citizen status, partner status, and dependent status. These variables describe the general background of the customer. They may help the model identify whether churn patterns differ between customer groups.

The service variables describe the products used by each customer. These include telephone service, multiple telephone lines, internet service, online security, online backup, device protection, technical support, streaming television, and streaming movies. Some service variables include values such as No internet service or No phone service. These values provide information about services that were not available to the customer.

The account variables describe the relationship between the customer and the company. Tenure shows the number of months that the customer has remained with the provider. Contract describes whether the customer has a monthly, one year, or two year agreement. Other account variables include paperless billing and payment method.

The financial variables are MonthlyCharges and TotalCharges. MonthlyCharges represents the amount charged to the customer each month. TotalCharges represents the total amount charged during the customer relationship. These values are important because payment amount and contract length may influence customer behaviour.

The customerID column provides a unique value for every customer. It is useful for connecting predictions and explanation results with individual customers. However, it

does not describe customer behaviour and was therefore not used as an input for model training.

The original dataset contains 19 usable customer features after removing customerID and the Churn target. During preprocessing, categorical variables were encoded into numerical form. This increased the number of model input columns to 45. During the explanation stage, the transformed columns were grouped back into the original 19 customer features so that the SHAP results could be presented using understandable business terms.

The TotalCharges column required additional attention. It appeared as a text based column because 11 customer records contained blank values. These values were converted into missing numerical values during data cleaning. The missing records were then handled before model training. All other columns were complete in the version of the dataset used for this thesis.

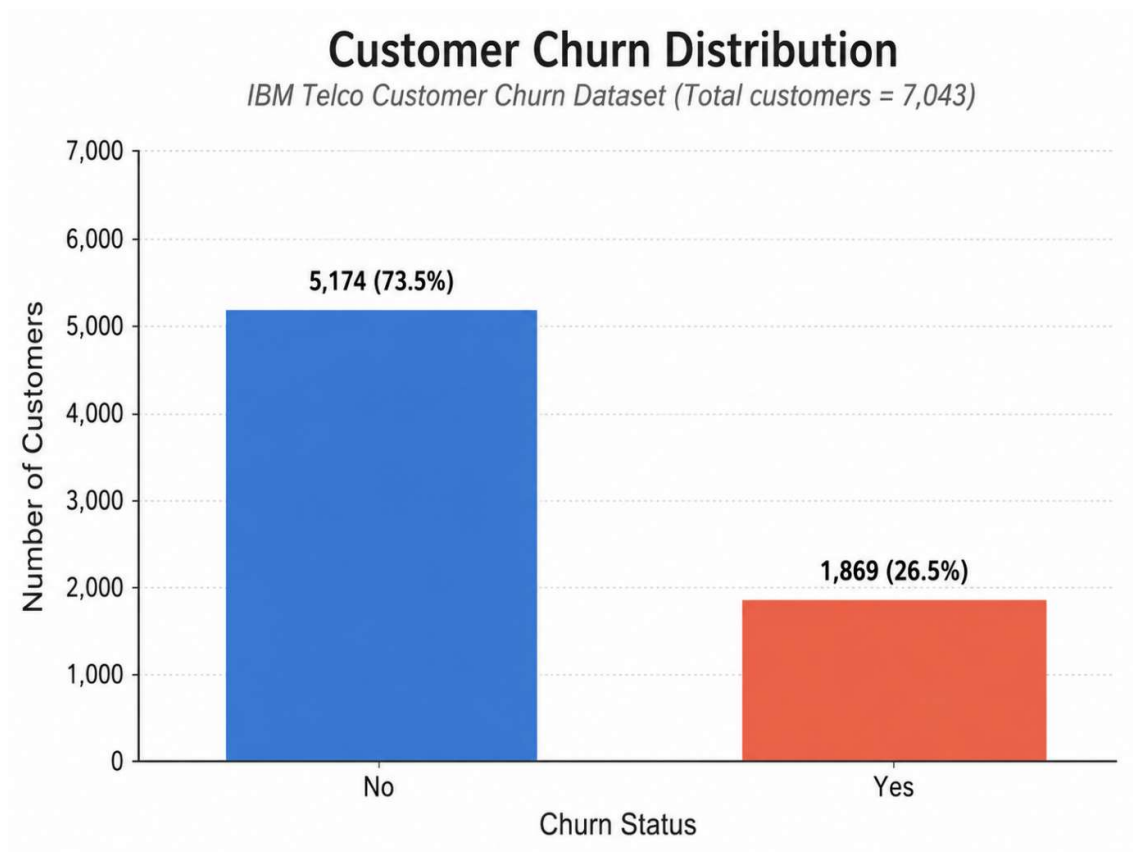
The dataset was suitable for the study because it contains a clear classification target and a mixture of demographic, service, contract, and financial information. It also supports both model development and business interpretation. However, it represents one fictional telecommunications case and does not include detailed customer behaviour, customer communication, competitor information, or retention actions. These limits are considered when discussing the results.

Table 5 Main variable groups in the Telco Customer Churn dataset

Variable group	Variables	Purpose
Customer identification	customerID	Identifies each customer
Demographic information	gender, SeniorCitizen, Partner, Dependents	Describes customer background
Service information	PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies	Describes subscribed services

Variable group	Variables	Purpose
Account information	tenure, Contract, PaperlessBilling, PaymentMethod	Describes the customer relationship and billing arrangement
Financial information	MonthlyCharges, TotalCharges	Describes customer charges
Target	Churn	Shows whether the customer left

Figure 3 Distribution of the customer churn target



3.3 Exploratory Data Analysis

After all the initial data cleaning has been performed, it was time to apply exploratory data analysis (EDA). This task involved inspecting the shape, distribution of individual variables, and some possible relationships among them which could be relevant for customer churn prediction.

All the analysis was performed using the Google Colab and Python language. Pandas and NumPy libraries were used for examining the data and calculating statistical values while Matplotlib and Seaborn libraries were used for visualization and comparing customer groups.

At the very beginning, the size of dataset and names and type of columns along with first few rows have been inspected. This confirmed that the cleaned dataset consisted of 7032 complete records and there were no remaining missing values.

Analysis of the churn target variable indicated that the outcome was not balanced. The dataset contained 5163 customers who did not churn and 1869 customers that did churn. Churned customers comprised 26.6% of the total clean data.

It is also quite important that this level of imbalance can influence model building and evaluation, so model should not treat only one class, assuming that is how most customers behave.

For model that predicts no churn, this would mean high overall accuracy although it misses a lot of actual churners.

Contract type vs Churn. A contract type means a type of engagement between a customer and the company. Three contract types were monthly, one year and two year which are analysed separately.

Tenure vs Churn. Tenure represents time of a customer since joining the company. The distribution was reviewed in relation with the outcome to see if customers with different life cycle with the company act differently.

Financial variables. Monthly Charges and Total Charges is analysed. Monthly Charges shows customer monthly bill while Total charges mean all money paid to the company by one customer during his lifetime.

Service-related analysis: Internet Service, Online Security, Online Backup, Device Protection, Technical Support, Streaming Television, and Streaming Movies analysis is conducted. The purpose is to investigate correlation with the churn.

Payment method vs Churn and Paperless Billing vs Churn is analysed as they are connected with how customers pay for the service they are getting from the company.

Demographic variables - gender, senior citizen, partner, and dependent. These are important features and were investigated against churn. Note that they cannot have a cause and effect in reality though they can lead into patterns which later can be explained.

Descriptive statistics (mean, std, min, max, quantiles) are calculated for Numerical variables - tenure, MonthlyCharges, and TotalCharges.

Correlation matrix between all Numerical variables were generated. As expected, strongest relations appear between Tenure and TotalCharges. Similarly, MonthlyCharges has relation with TotalCharges as expected. No other significant numerical relations appeared unexpected or unknown.

Outlier Detection for numerical variables (tenure, MonthlyCharges and TotalCharges) was performed using Interquartile Range (IQR) method to remove unusual values, as expected and found that there were not unusual values and none needs further processing in terms of outlier's removal.

Following is a summary of the important variables that are selected after the initial EDA. The EDA also proved that the dataset is well suited for further analysis as well as for the development of the customer churn prediction model and its subsequent interpretation.

Figure 4 Correlation matrix of the numerical variables

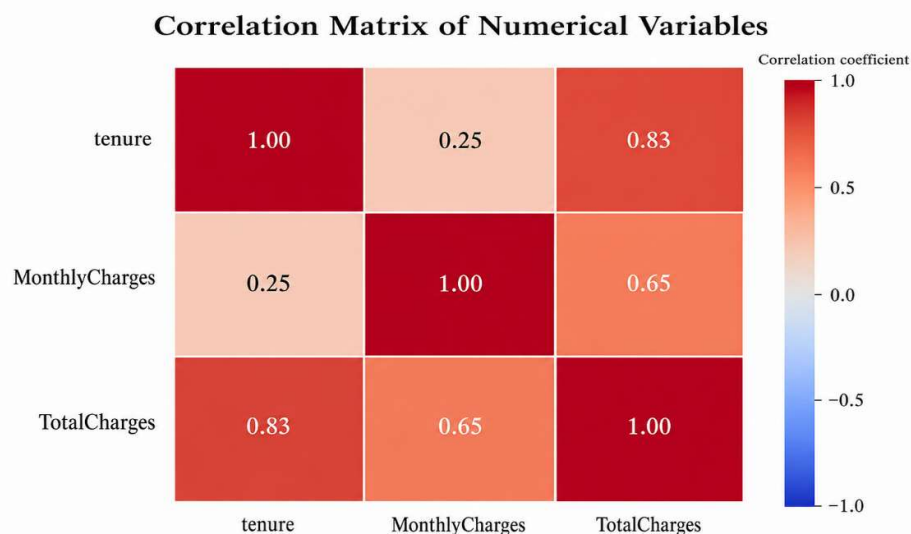


Table 6 Exploratory analyses completed for the customer churn dataset

Analysis area	Variables examined	Purpose
Target distribution	Churn	Examine class balance
Contract analysis	Contract and churn	Compare churn across contract types
Customer relationship	tenure and Churn	Examine churn across customer duration
Financial analysis	MonthlyCharges and TotalCharges	Examine customer charges
Service analysis	Internet and support services	Compare churn across services
Payment analysis	PaymentMethod and PaperlessBilling	Examine billing related patterns
Demographic analysis	gender, SeniorCitizen, Partner, Dependents	Compare customer groups
Correlation analysis	Numerical variables	Examine relationships between values
Outlier analysis	tenure, MonthlyCharges, TotalCharges	Identify extreme numerical values

3.4 Data Cleaning

Data cleaning was done prior to the machine learning preprocessing stage. The cleaning was done to address data quality concerns, and create a complete data frame, which could be used later on in the project pipeline. Initializing a dictionary for and imputing the missing values 7,043 customer records and 21 columns are contained in the original data frame.

After the data frame was imported to Google Colab, the column names, data type of columns, presence of missing values, and numerical summarizations of the data frames was analysed.

For the TotalCharges data column, there was an anomaly as although the TotalCharges is a numerical data value, it was classified and recorded as a text data type, which was then type-casted into a numerical column type using pandas data-frame functionality.

In the 21-column list, 11 of the records of customers had values of “blank” in TotalCharges which denotes missing values in the column which was type casted. As only 11 records in comparison to 7043 were missing in TotalCharges; rather than filling these missing values with estimated means it was preferred to remove the rows containing blank values as is recommended when the missing values amount to about 0.16 % of the original records. Thus, a cleaner data frame comprising of 7,032 customers were the result of this operation; A check done after removal of the abovementioned 11 record proved that there are no longer any missing values left in the data frame.

These same operations on the above-mentioned columns were conducted in the preprocessing column of the notebook, so that it did not contain decencies on the manually cleaned data from Exploratory analysis part.

So, the notebook on preprocessing may be utilized as a standalone. The data frame contained customerID and Churn as separate attributes in 7,032 rows. The customerID attributes have been left for later use so that, it helps in identifying the individual customer, SHAP values, and in Power BI to relate the prediction to the corresponding individual.

However, it may not be of any help in the machine learning model to predict on customer behaviour, due to its identity nature. Also, the churn values for the identified customer will be used as the target variable. The remaining columns were divided into numerical type and categorical type of columns for use in later stage in the preprocessing column.

Some of the categorical attributes were also analysed for unusual values; no obvious significant outliers were discovered in any of the numerical columns (tenure, MonthlyCharges, and TotalCharges) that warrants removing any additional rows due to large values of attributes.

Thus, on completion of these procedures, the data was clean of missing values and contains appropriate numerical value in the columns which are relevant to the prediction.

Table 7 Summary of the data cleaning process

Cleaning step	Result
Original dataset size	7,043 customer records
TotalCharges data type	Converted from text to numerical format
Missing TotalCharges values	11 records
Missing value treatment	Incomplete records removed
Final cleaned dataset	7,032 customer records
Remaining missing values	None
Identifier handling	customerID preserved but excluded from model inputs

3.5 Data Preprocessing

After the data was tidied up, the next step of the ML process was to prepare the data for the model, to convert the raw variables into a numerical and consistent format, suitable for different classification models. Churn was selected as the target variable, and the existing categorical 'No' and 'Yes' values were converted into '0' and '1' (no churn and churn respectively). After this conversion the target variables would contain 5,163 non-churn and 1,869 churn customers.

The customer ID feature was removed as this only represents and identifies a customer but does not represent their behaviour, but this customer identifier has been saved as a reference, and will be useful to associate predictions and explanations to customers later on.

The other 19 input features can be grouped into 3 categories: Continuous numerical, Binary numerical, and Categorical: Three features were classified as continuous numerical; tenure, MonthlyCharges, and TotalCharges. Only Senior Citizen was classified as a binary numerical feature, with the remaining 15 features being classified as categorical. The complete dataset was split into the training and testing datasets, with 80% of rows being allocated to train the models and the remaining 20% reserved for the testing dataset. A random state number of 42 has been used so that the split could be reproduced.

Stratification has been applied with the target variable in order to maintain a balance of churn and no-churn customers in both the training and testing datasets, the training dataset contained 5,625 customers, 1495 churn and 4,130 non-churns, while the testing dataset contained 1,407 customers, 374 churn and 1033 non-churn.

As the continuous numerical variables, such as tenure and the charges had vastly different ranges, these were standardised using a scikit-learns, StandardScaler, which was fit on the training data only, which will help scale the values of the training data to a mean near zero and a standard deviation of one, this then when applied to the testing data the results are on the same scale. Senior Citizen was not standardised as this feature is already represented by 0 and 1 values. Categorical variables were converted to a numerical representation using scikit-learns one hot encoder, where each categorical value will have its own associated column of 0s and 1s, a strategy has also been applied for 'unknown' categories in case new categorical values in the testing or production set that have not been seen in the training set were present, these are ignored in that scenario.

To ensure consistency the various transformations were applied together in a single scikit-learns Column Transformer, this would help the transfer of processing steps across the training and testing sets with no variations between them; standardisation was applied to the continuous numerical, senior citizen remained untouched and the one-hot encoding of all categorical variables. Only the training data was used to 'fit' the preprocessing transformer, before being used to 'transform' both the training and testing data. This prevents data from the testing set influencing the pre-processing which may lead to an overestimation of model performance.

As a result of pre-processing, the original 19 input variables were transformed into 45 model input features.

The processed training data contained 5625 rows and 45 columns of data; the processed testing data contained 1407 rows and the same 45 columns of features. Data validation checks were made including that both the processed datasets contained no missing or infinite values, no columns were unique for either training and testing sets, all model inputs are numerical and the churn values correctly match the customer rows they are associated with. Training and testing 'reference' data files also were created containing customer ID, all original feature values, original target variable values, and processed target variables, these will be used to relate all prediction values back to customers (e.g., predictions, churn probabilities, SHAP explanation values). The complete pre-processing stage as well as the data validation checks can be seen in more detail within the phase 3 final report, and has resulted in training and testing datasets that are ready to be fed into ML models.

Figure 5 Data preprocessing pipeline used in the thesis

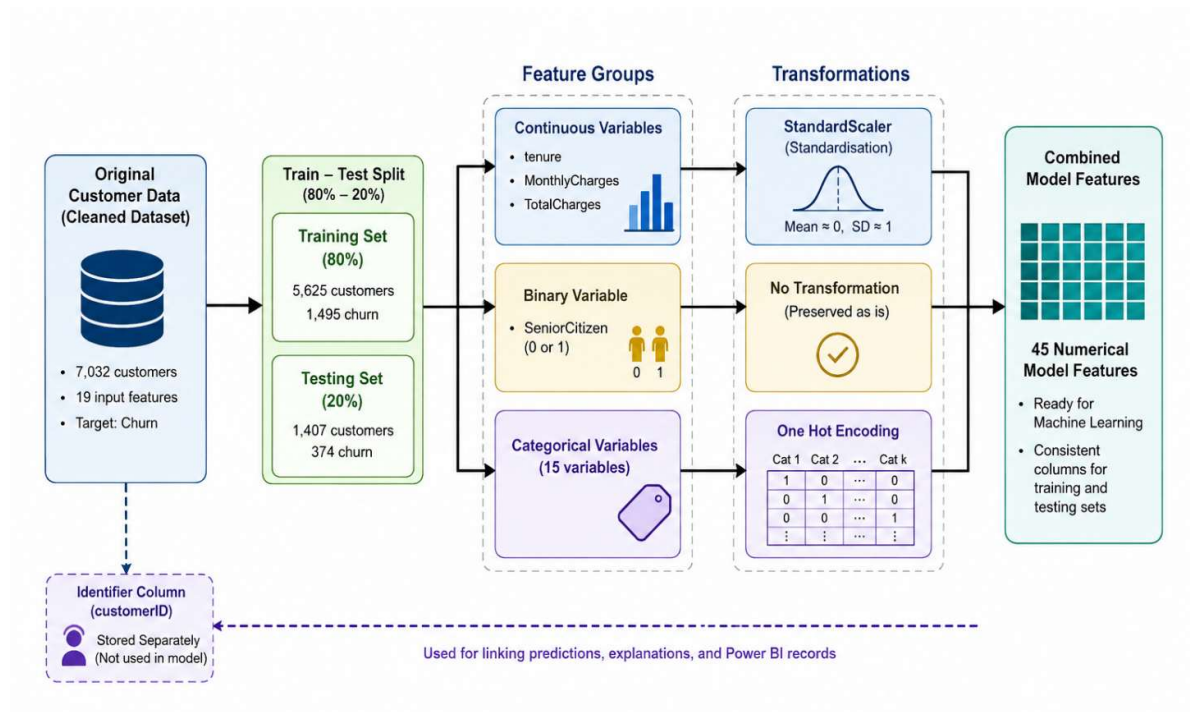


Table 8 Summary of the preprocessing transformations

Feature group	Variables	Transformation
Continuous numerical	tenure, MonthlyCharges, TotalCharges	Standardisation
Binary numerical	SeniorCitizen	Preserved without transformation
Categorical	15 categorical variables	One hot encoding
Identifier	customerID	Removed from model input and stored separately
Target	Churn	No encoded as 0 and yes encoded as 1

3.6 Training and Testing Strategy

Before applying the preprocessing transformer, the already cleaned dataset was split into training and testing data. In the end we need to assess how the model will perform using customers that were not used while building the model so for that reason it is important to keep the testing dataset apart. The split was defined 80% for the training, remaining 20% for the testing.

The total customers of the dataset are 7,032 since 11 records were removed with unknown value.

There are 5,625 customers in train dataset with split to classes being 1,495 churn and 4,130 actives (still). There is 1,407 in test dataset with 374 being labelled as churn customers. We've used random state value 42, in that case no matter how many times will we rerun the code, the customer's position in train/test group stays the same. It is crucial that the random state is chosen because it guarantees that the models can be comparable for different reasons.

The split was performed using "Stratify by Churn target", this technique allows to guarantee that both the training and the test set maintain a similar percentage between the target classes.

So, without stratification there could be a noticeable difference for the churn percentage between the two sets. The training dataset was used for preprocessing, for training of the model and finally for the selection and tune of the model and parameters; on the other hand, test set were kept hidden. For that reason, the model will not adapt and learn about that specific customer data.

In order to not bring any data from the test set to the training process, the test set will not be used to compute any mean, std or category for scaling and encoding of data, but the train set. It is necessary because otherwise, our evaluation measures could report values higher than they actually are. The test dataset was used to compute final evaluation measures; it is included accuracy, precision, recall, F1-score, ROC AUC, PR AUC, Brier Score and the confusion matrix of the predictions made on the testing set and not the entire population of customers which is for example used in the calculation of the final evaluation measurements in the other notebooks which is the confusion matrix or the ROC curve etc.

The testing dataset was also used to generate the final customer predictions, churn probabilities, and SHAP explanations that were prepared for Power BI. Keeping this dataset separate provided a more realistic estimate of model performance on unseen customers. However, the results are based on one fixed split of one public dataset, so

the same performance cannot be guaranteed for every future dataset or telecommunications company.

Table 9 Distribution of customers in the training and testing datasets

Dataset	Total customers	Churn customers	Non churn customers
Training	5,625	1,495	4,130
Testing	1,407	374	1,033
Complete cleaned dataset	7,032	1,869	5,163

3.7 Machine Learning Models

During the model development all the five classification models were considered. These were Logistic Regression, Weighted Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting. The models were selected to compare different approaches to binary classification. This allowed simpler models to be compared with more complex tree-based models.

Logistic Regression was used as the main baseline model. It estimates the probability that a customer belongs to the churn class by learning a relationship between the input features and the target variable. The model is relatively simple, and its probability output can be used directly for customer risk analysis [10], [11].

Weighted Logistic Regression uses the same general approach as Logistic Regression but gives greater importance to the smaller churn class during training. The purpose of using this model was to reduce the effect of class imbalance and identify a larger proportion of customers who actually churned.

A Decision Tree was included because it can learn nonlinear relationships by dividing the customer data through a series of conditions. The depth of the tree and the minimum number of customers required in the final groups were controlled to reduce excessive model complexity [11].

The decision tree results are taken together by the random forest. Each tree is trained using a random selection of customer records and input features. The predictions

produced by the trees are then combined to obtain the final result. This method can reduce the instability of a single Decision Tree, although the complete model is more difficult to interpret directly [27].

Gradient Boosting also combines several Decision Trees, but the trees are created in sequence. Each new tree attempts to correct some of the errors made by the earlier trees. This allows the model to improve gradually and learn more complex relationships within the customer data [28].

All five models were evaluated using five fold stratified cross validation on the training dataset. The training records were divided into five groups with similar churn proportions. During each round, four groups were used to train the model and the remaining group was used for validation. This process was repeated until every group had been used once for validation. The independent testing dataset was not used during this comparison.

The models were compared using several evaluation measures instead of accuracy alone. This was important because churn customers formed the smaller class in the dataset. The comparison included precision, recall, F1 score, ROC AUC, Precision Recall AUC, probability quality, and the difference between training and validation performance.

The first comparison showed clear differences between the models. Best Churn Recall with Weighted Logistic Regression 0.8027, Decision tree identified lots of churn customers 0.7967. Best Base Probability ranking result with Gradient Boosting - ROC AUC 0.8494, Precision recall AUC 0.6683, it worked better for ranking churn customers over non-churn customers among other base models. Random forest performance on validation set is acceptable, but there is a larger gap between train and validation performance as F1 score was approximately 0.1841, which implies higher potential overfitting.

Then, parameters for Logistic Regression and Gradient Boosting model were tuned, we used only the training dataset.

Gradient Boosting was finally selected as our final model due to its excellent cross-validation score (ROC AUC and Precision Recall AUC), consistency across different groups, lower overfitting risk, and suitability for Tree based SHAP explaining approach.

Table 10 Machine learning models evaluated in the study

Model	Main purpose in the comparison	Main characteristic
Logistic Regression	Provide a simple baseline	Produces class probabilities using a linear relationship
Weighted Logistic Regression	Address the smaller churn class	Gives more importance to churn customers during training
Decision Tree	Examine a single nonlinear tree model	Divides customers using a sequence of conditions
Random Forest	Examine an ensemble of independent trees	Combines predictions from many randomised trees
Gradient Boosting	Examine a sequential tree ensemble	Each new tree improves earlier prediction errors

3.8 Model Evaluation Metrics

Several evaluation measures were used to compare the classification models. Accuracy alone was not considered sufficient because the dataset contained more non churn customers than churn customers. A model could achieve reasonable accuracy by predicting the larger class correctly while failing to identify many customers who actually churned.

The positive class was defined as Churn equal to 1. Therefore, the evaluation focused particularly on the ability of each model to identify actual churn customers. For this purpose, we measured the metrics like accuracy, precision, recall, F1 score, ROC AUC, Precision Recall AUC, Brier score, and confusion matrix values. These measures were calculated during cross validation and during the final evaluation on the independent testing dataset.

Accuracy represents the proportion of all customers who were classified correctly. It includes both correctly identified churn customers and correctly identified non churn customers. Although accuracy provides a general summary of performance, it does not show how well the smaller churn class was identified.

Precision measures the proportion of customers predicted as churn who actually churned. Higher precision indicates that fewer non churn customers were incorrectly classified as churn customers. This is important because incorrect churn predictions could lead to retention actions being directed towards customers who were not expected to leave.

Recall: The ratio of churn customers correctly classified. This metric was of particular importance as a false negative represents a customer who was predicted to stay, and instead churned. Such customers would not be identified for possible retention actions.

The F1 score combines precision and recall into one measure. It is used for assessment balance between correctness of predicting churn customers, and minimization incorrect churn predictions. Bigger result from F1 Score means both measures are being balance in best condition [11].

ROC AUC evaluates how effectively a model ranks churn customers above non churn customers across different classification thresholds. A value closer to one indicates stronger separation between the two classes, while a value near 0.5 indicates performance similar to random ranking [29].

Precision Recall AUC summarises the relationship between precision and recalls across different thresholds. It was included because the churn class was smaller than the non-churn class. Precision Recall analysis can provide useful information about positive class performance when the target distribution is imbalanced [30].

The Brier score assesses the gap between observed churn values and the forecasted churn probabilities; it gets lower when the forecasted probabilities move closer to the observation. This measure was included because the final dashboard displayed customer churn probabilities rather than only predicted class labels [31].

A confusion matrix was used to examine the number of true positives, true negatives, false positives, and false negatives. A true positive was an actual churn customer correctly predicted as churn. A true negative was a non-churn customer correctly predicted as non-churn.

A false positive was a non-churn customer incorrectly predicted as churn. A false negative was an actual churn customer incorrectly predicted as non-churn. The false negative count was given particular attention because it represented churn customers missed by the model.

The difference between training and validation performance was also examined. A large disparity may suggest a model had “overfitted to the training data” and therefore “may not perform as well on customers the model had never seen previously”. This

comparison was therefore used as an additional indication of model stability and possible overfitting.

Using several evaluation measures made it possible to examine classification accuracy, churn detection, probability ranking, probability quality, and model stability. The final model was not selected according to one measure alone. Instead, the complete set of results was considered together with the research objective and the requirements of the later SHAP and Power BI stages.

Table 11 Evaluation metrics used for model comparison

Metric	Main purpose	Preferred direction
Accuracy	Measures the proportion of all correct predictions	Higher
Precision	Measures how many predicted churn customers actually churned	Higher
Recall	Measures how many actual churn customers were identified	Higher
F1 score	Balances precision and recall	Higher
ROC AUC	Measures ranking across classification thresholds	Higher
Precision Recall AUC	Measures positive class performance across thresholds	Higher
Brier score	Measures the quality of predicted probabilities	Lower
Confusion matrix	Shows correct and incorrect prediction counts	Depends on error type

3.9 Model Selection

The final model was selected through a comparison of predictive performance, model stability, probability ranking, and suitability for the later explainability and dashboard stages. The independent testing dataset was not used during model tuning or model selection.

The baseline comparison showed that Logistic Regression and Gradient Boosting produced strong probability ranking results. Therefore, these two model families were selected for limited parameter tuning using only the training dataset.

For Logistic Regression, different regularisation strengths and class weight settings were examined. For Gradient Boosting, the tuning process considered the number of estimators, learning rate, maximum tree depth, minimum samples per leaf, and subsample size.

The optimal settings for Gradient Boosting parameters were: number of estimators: 100, learning rate: 0.05, max depth of tree: 2, min samples in a leaf: 30 and subsample = 0.80. The use of shallow trees, a low learning rate, and a minimum leaf size helped control the complexity of the model.

The two tuned candidates were compared using five fold stratified cross validation. Tuned Gradient Boosting achieved an average accuracy of 0.8057, precision of 0.6837, recall of 0.5023, F1 score of 0.5786, ROC AUC of 0.8496, and Precision Recall AUC of 0.6693.

Tuned Logistic Regression achieved an average accuracy of 0.8025, precision of 0.6538, recall of 0.5472, F1 score of 0.5955, ROC AUC of 0.8461, and Precision Recall AUC of 0.6601.

Tuned Logistic Regression produced a slightly higher F1 score at the standard classification threshold. The Tuned Gradient Boosting produced the better results for ROC AUC and Precision Recall AUC. These metrics were selected because the model needed to be able to order the customers by their prediction of chance of churning.

The selection was made considering model stability too, in that the Tuned Gradient Boosting produced similar results across the validation folds and that the F1 score difference between training performance and validation performance was around 0.0165 which indicates that there is little danger of significant over fitting of the model.

The selected model was the Tuned Gradient Boosting Classifier. The justification for this was that it produced the best probability ranking performance, produces reliable, stable validation performance figures, produces customer level probabilities and allows tree based SHAP explanations to be provided. It could also then be readily exported together with customer predictions and SHAP explanations for use within Power BI.

Once model selection was finalised, the final Gradient Boosting model was built using the entire set of 5,625 training records. None of the independent test dataset were able to be used till the last section to avoid a selection bias in the accuracy measuring

of the final model and the result set and all related configuration are supported in phase 4 documentation.

Automatic calculation of a threshold from the test set was avoided and was separately identified during a threshold analysis using predictions produced using the training set.

Table 12 Comparison of the tuned candidate models

Model	Accuracy	Precision	Recall	F1 score	ROC AUC	Precision Recall AUC
Tuned Gradient Boosting	0.8057	0.6837	0.5023	0.5786	0.8496	0.6693
Tuned Logistic Regression	0.8025	0.6538	0.5472	0.5955	0.8461	0.6601

3.10 Decision Threshold Selection

The Gradient Boosting Model selected determined a churn probability for each individual customer. We needed to set a classification threshold to turn our churn probability into a final churn prediction.

Our initial, default classification threshold was set to 0.50. Using this threshold, any customers with a probability of 0.50 or higher was predicted as churn and all other customers were predicted as non-churn.

The default threshold gave us reasonable overall accuracy, but we only detected around half of our actual churners. This clearly did not meet our model requirements since anyone classified as non-churn wouldn't be considered for any potential retention actions.

Therefore, we needed to explore different thresholds based on out-of-fold churn probabilities obtained from the training dataset. We are training customer probability on models that were not trained using that customer. This was a much more realistic approach to training threshold selection than using probabilities drawn from the trained training model itself.

The separate independent testing data was not used at this point of exploration to prevent influencing the actual threshold used for final test evaluation.

Our goal was to achieve a target churn recall of at least 80%. This thresholds exploration will balance Accuracy, Precision, Recall and F1, false positives, and false negatives. Lowering the thresholds significantly increased recall, but also the number of false positives (incorrect predictions of churn).

Under the threshold of 0.50 (default):

Out-of-fold Training Predictions

Accuracy:0.8057

Precision:0.6827

Recall:0.5023

F1 Score:0.5788

False Positives: 349

False Negatives: 744 (missed actual churners)

Under the threshold of 0.27 (chosen):

Out-of-fold Training Predictions

Accuracy:0.7579

Precision:0.5293

Recall:0.8027

F1 Score:0.6380

False Positives: 1,067

False Negatives: 295

So, by lowering the threshold from 0.50 to 0.27 we increased churn recall from around 50.2% to 80.3% and increased the F1-score to around 63.8%. Unfortunately, this increased churn detection comes with decreased accuracy and precision. We were happy to take this trade-off however, because for our objective missing a true churn customer was more undesirable than incorrectly identifying some non-churn customers. If a customer is wrongly predicted as high-churn, we may lose a small cost by sending them an unnecessary retention communication; if, however we miss a true churn customer, they simply get no retention message.

We will store the 0.27 thresholds with the trained model and apply it during our independent test evaluation which will be presented in Chapter 4.

Table 13 Comparison of prediction thresholds using training out of fold results

Metric	Default threshold 0.50	Selected threshold 0.27
Accuracy	0.8057	0.7579
Precision	0.6827	0.5293
Recall	0.5023	0.8027
F1 score	0.5788	0.6380
False positives	349	1,067
False negatives	744	295

3.11 SHAP Explanation Method

After the final Gradient Boosting model had been trained and evaluated, SHAP was used to explain its predictions. The purpose of this stage was to show how each customer feature influenced the predicted churn probability. SHAP supports both global explanations of overall model behaviour and individual explanations for specific customers [2].

The saved Gradient Boosting model was loaded together with the processed training and testing datasets. The saved feature names, customer reference records, churn probabilities, and model metadata were also loaded. Before generating explanations, the feature order and customer IDs were checked to ensure that they remained aligned with the model outputs.

The model explained the positive target class, where Churn was equal to 1. The selected prediction threshold remained 0.27. However, the threshold was used only to convert churn probabilities into predicted churn labels. It did not change the SHAP contribution values because SHAP explained the underlying probability produced by the model.

SHAP TreeExplainer was selected because the final model was based on decision trees. The explainer used the interventional feature perturbation method and generated

explanations directly on the probability scale. This made the contributions easier to connect with the churn probabilities displayed in Power BI.

A reproducible background sample of 100 customers was selected from the training dataset using a random state of 42. This background data provided a reference point for calculating how each customer feature moved the prediction away from the average model output.

SHAP values were calculated for all 1,407 customers in the independent testing dataset. Each customer initially received one contribution value for each of the 45 processed model features. A positive SHAP value indicated that a feature increased the predicted churn probability relative to the base value. A negative SHAP value indicated that a feature reduced the predicted churn probability.

The base value represented the model's expected churn probability based on the background dataset. For each customer, the base value and all feature contributions were combined to reconstruct the customer's predicted churn probability. An additivity check was performed for every customer to confirm that the explanation values correctly matched the model output.

The 45 processed model features included standardised numerical variables, the SeniorCitizen variable, and the columns created through one hot encoding. These processed features were useful for the model but were not always easy for a business user to interpret.

A feature mapping was therefore created to connect every processed feature with its original customer variable. For example, the different encoded Contract columns were grouped under the original Contract feature. This process converted the 45 processed feature contributions into contributions for the 19 original customer features.

The grouped contribution for an original feature was calculated by adding the SHAP values of all processed columns connected to that feature. A validation check confirmed that the total grouped contribution remained equal to the total contribution from the processed features.

Global feature importance was calculated using the average absolute SHAP value of each feature across all testing customers. Absolute values were used because global importance measures the strength of a feature's influence regardless of whether it increased or reduced churn probability.

Individual customer explanations were also created. For each customer, the strongest positive factors and strongest negative factors were identified. These results made it

possible to explain which customer characteristics increased churn risk and which characteristics reduced it.

The outputs of SHAP have been extracted into both wide and long format files. In the outputs we can find global importance value of features for model and customer, individual contribution of each feature to prediction for each customer, direction of each individual contribution, values of the customer features, sum of the biggest positives and negatives explanations.

Validation checks proved that each of the 1407 test customers are assigned a SHAP explanation for all of the 45 processed features. Validation included missing and infinite value check, feature mapping check, customer matching check, output dimension and grouping of the contributions as well as additivity check. Processed features have been grouped correctly to the 19 original customer features.

With this approach, all layers of the explanation were created, and it is possible to integrate them into interactive visual analytics tool. Model generates churn probabilities, SHAP explains the dependency of these predictions from customer variables, and these two pieces will be visualised together with use of Power BI.

Table 14 SHAP explainer configuration used in the study

Configuration item	Selected setting
Explanation method	SHAP TreeExplainer
Final model	Tuned Gradient Boosting Classifier
Explained class	Churn equal to 1
Feature perturbation	Interventional
Model output	Probability
Background dataset	100 training customers
Random state	42
Customers explained	1,407 testing customers
Processed features	45

Configuration item	Selected setting
Grouped original features	19

3.12 Dashboard Design Method

Once the prediction results and SHAP outputs were generated, they were then loaded and visualized using Microsoft Power BI Desktop into an interactive visual analytics system. A key design goal of the dashboard was to allow for user exploration of model results in a consumable form, without the need to interact directly with Python or any other machine learning code. Power BI was used for building, not for training a model or computing SHAP values.

These steps were performed using Python, and Power BI was used to connect prepared data tables, calculate reporting measures, produce visualizations, and enable interaction between dashboard elements.

The data tables imported into Power BI consisted of original customer variables, actual churn labels, predicted churn values, predicted churn probabilities, the selected threshold, global SHAP feature importances, individual customer SHAP values, and the strongest positive contributing factor and strongest negative contributing factor for each customer. Information relating to the model performance results from the previous phase was also imported. Since the customer prediction table contained one row for each of the 1,407 test customers, this table was used as the main customer table to ensure that unique customers would only appear once. The other tables represented individual customer explanations and therefore contained multiple rows for each customer. These records were intentionally preserved because each row represented a different feature and were connected to the main customer table through a one-to-many relationship.

Relationships were set based on customerID to enable a filter on one chosen customer to affect the related prediction and explanation information.

It was important to check the relationships to avoid duplication in counts of customers, as customer counts and churn figures were derived from the customer prediction table, not directly from the very long form SHAP table, which contained 19 records per customer. The DAX measures were built to drive the core dashboard components, such as total customers, customers with true churn, customers with predicted churn,

average prediction probability for the test set, customers with high predicted churn, as well as performance measures such as accuracy, precision, recall, F1 scores and confusion matrix component counts. This was supported by a visual environment consisting of six distinct pages all designed to facilitate the answers to a different category of questions, but each of them leveraging the same set of underlying prediction and explanation data.

The "Executive Overview" page provided a snapshot view of the test customer set with the most important indicators like total customers, true and predicted churn, average churn probabilities, and predicted customer risk levels. The "Customer Risk Analysis" page allowed customers to segment and filter the customer base by their predicted churn probabilities and predicted risk levels, such as investigating whether those with high predicted churn also demonstrated high actual churn behaviour. The "Global Model Explanation" page presented the global SHAP value results indicating the most influential customer characteristics that generally affect the model predictions across the entire test set.

The "Individual Customer Explanation" page offered an opportunity to examine a particular customer, such as customer #8, and get a detailed analysis on what attributes and to what extent were the primary contributors to their specific predicted churn value and probability. That was illustrated by showing their greatest positive influence, their greatest negative influence and the nature and magnitude of influence from the remaining features.

The "Customer Segment Comparison" page provided the capability to compare predicted churn metrics for specific segments and groups of customers categorized by common features such as contract type, internet service, payment method and tenure groups. The "Model Performance Summary" page presented the final classification results based on the identified and selected classification threshold, including the components of the confusion matrix (true positives, false positives, true negatives, false negatives), classification threshold, accuracy, recall, precision, and F1 score to see the impact of the 0.27 threshold over the standard 0.50 threshold. All of the visuals were connected, through extensive use of slicers and filters for customers and segments, to ensure that an interaction on one item (table, chart, card etc) triggered updates to the rest on the page. The user also had access to navigation controls that allowed for easy transitions between the six pages.

Maintaining a high level of consistency was achieved throughout the report by using consistent page design, fonts, titles and button positioning on each page.

The visuals on a single page were kept to a minimum, in an effort to reduce the cognitive overload for the user. Care was taken in how the SHAP contributions were described; rather than stating that the feature caused the customer to churn, it was phrased such that the feature increased the predicted churn probability, or decreased the predicted churn probability. This is due to SHAP's methodological limitation in establishing causal relationships. SHAP explains feature influences on a trained model's predictions rather than proving what caused the real outcome.

The completed system was extensively validated by inspecting relationships, customer counts, DAX measures, slicer functionality, interactions, prediction values, threshold levels, and SHAP influence directions against the source data. Several individual customer predictions from the dashboard were compared directly with their corresponding records in the original Python output files to verify accuracy. The system presented a pathway from a high-level prediction of customer churn risk to an explained prediction for an individual customer.

The combined analysis allowed the machine learning prediction, SHAP explanation, and business visualization stages to be connected within a single interactive application. This multi-page architecture and interaction setup were used later during the comparison with the static explanation baseline.

Table 15 Purpose of the Power BI dashboard pages

Dashboard page	Main purpose
Executive Overview	Present the main customer, churn, probability, and risk indicators
Customer Risk Analysis	Identify and examine customers across different churn risk levels
Global Model Explanation	Present the most influential model features using global SHAP importance
Individual Customer Explanation	Explain the prediction and SHAP contributions for one selected customer
Customer Segment Comparison	Compare churn patterns and risk between customer groups

Dashboard page	Main purpose
Model Performance Summary	Present final evaluation results, threshold information, and confusion matrix values

3.13 User Evaluation Method

A user evaluation was conducted to compare the static explanation baseline with the interactive Power BI dashboard. The objective was to examine whether dashboard interaction improved users' understanding of model predictions, their ability to identify important churn factors, and their ability to make customer retention decisions more effectively.

Both the static baseline and the interactive Power BI dashboard presented information from the same machine learning model, customer data, predicted churn probabilities, SHAP explanations, and model performance measures. The main difference between the two systems was that how the information was presented. So first the static baseline displayed screenshots and summaries in a fixed format, while on the other hand the interactive Power BI dashboard allowed users to apply filters, select specific customers, compare segments, and navigate through six report pages.

A within subject design was used to compare the two systems. Each participant evaluated both systems, which allowed their performance under the two conditions to be compared directly. This approach also reduced differences caused by individual abilities, backgrounds, and previous experience.

A total of 10 participants from different backgrounds took part in the user evaluation. Their fields of study consisted of data science, cybersecurity, aviation, architecture, and physics. Their previous experience with Power BI ranged from no experience to an intermediate or basic level of knowledge. Participant identification codes from P01 to P10 were used during the analysis instead of individual names.

The order of the two systems was counterbalanced to reduce the influence of learning and presentation order. Participants P01 to P05 evaluated the static explanation first and the interactive dashboard second. Participants P06 to P10 completed the evaluation in the opposite order.

The 10 participants completed a set of tasks using both systems. These tasks measured global model interpretation, individual customer explanation, customer risk analysis, customer segment comparison, and model performance interpretation.

The tasks required participants to identify the most important global churn feature, locate customers with high predicted churn risk, and identify the factors that influenced the prediction for a selected customer. Participants also compared customer segments and interpreted the selected classification threshold and model performance measures.

Task accuracy was measured by comparing participant responses with the correct answers obtained from the model and dashboard outputs. Completion time was recorded for each system. The number of tasks completed correctly was also calculated separately for the static and interactive conditions.

After completing the tasks for each system all the participants provided ratings for understanding, trust in the model and explanation, decision support, usability, and perceived usefulness. A rating scale from 1 to 5 was used, where higher values were representing a more positive outcome.

Understanding referred to how effectively participants could interpret the churn probability, predicted class, SHAP contributions, global model behaviour, and model performance.

Trust referred to the credibility and clarity of the model prediction and explanation. Decision support measured whether the information helped participants decide which customers should be considered for retention actions. Usability referred to how easily the systems could be operated and navigated. Perceived usefulness represented the value participants believed the systems provided for customer churn analysis.

Participants were then asked to select their preferred system and provide a short explanation for their choice. Their preferences and written explanations were included in the final evaluation results.

Each evaluation session followed the same general procedure. Participants first received a short introduction explaining the evaluation and the assigned system. They then completed the tasks while their responses and completion times were recorded. A rating session followed the task section.

The same procedure was repeated for the second system. After evaluating both systems, participants selected their preferred system and provided a short written reason. Each complete evaluation session required approximately 15 to 20 minutes.

Data collected: codes for participant identification, background information of participants, previous Power BI experience of participant, sequence of systems and its order; response to each task; number of errors made; time consumed for each task; participant rating for system usability and reasons for user preference of system, and for not using a particular system.

The quantitative analysis compared the mean task accuracy, mean completion time, and average rating scores of the two systems. The number of participants who preferred each system was also calculated. Written feedback was analysed and organised to identify common positive points, difficulties, and suggestions.

Because only 10 participants took part in the study, the findings were treated as evidence from a small practical evaluation. The results cannot be broadly generalised to every possible business user. However, the main purpose was to examine whether the interactive dashboard provided practical advantages over a static explanation format for understanding customer churn predictions.

The participant order, evaluation measures, task sequence, and analysis method were defined before the final results were processed. The numerical and qualitative evaluation findings are presented in the results chapter.

Table 16 Summary of the user evaluation design

Evaluation element	Method used
Participants	10 participants with different backgrounds
Study design	Within subject comparison
Conditions	Static explanation baseline and interactive Power BI dashboard
System order	Five participants used static first and five used interactive first
Task areas	Global explanation, individual explanation, risk analysis, segment comparison, and model performance
Performance measures	Task accuracy and completion time
Rating measures	Understanding, trust, decision support, usability, and perceived usefulness
Rating scale	Five points from strongly disagree to strongly agree
Final response	System preference and short written feedback
Approximate duration	15 to 20 minutes per participant

3.14 Ethical Considerations

Ethical issues have been taken into account during the data preparation, the model building, the explanation, the dashboard as well as the user evaluation. Top prioritized issues are Data Privacy, Participants confidentiality, responsible model interpretation of predictions, Fairness, Transparency, and automated decision support.

The Telco Customer Churn is a public data sample for analytic purpose and doesn't store any customer names, Addresses, or any specific contact information for individuals. The customerID is a private key used solely for matching the customers between the data table, prediction results and the SHAP values at each step of the projects.

The customerID field was excluded from the machine learning input because it did not describe customer behaviour. It was stored separately only to maintain the correct connection between customer records and analytical outputs.

The user evaluation involved information about participants, including their educational or professional background and previous Power BI experience. Participant identification codes from P01 to P10 were used during analysis and reporting instead of individual names.

Participant names and other identifying information should not be included in the publicly available thesis files, appendices, or GitHub repository. The public evaluation results should contain only participant codes and combined summary results.

The model was developed as a decision support tool rather than an automatic decision making system. A prediction that a customer may churn should not automatically lead to a business action without further review. The model provides estimated probabilities based on historical patterns and cannot know every reason behind an individual customer's behaviour.

The selected threshold of 0.27 increased the number of churn customers identified by the model. However, this also increased the number of false positive predictions. False positive leads to retention attention directed towards a non churning customer, a false negative gives no retention attention to a churning customer.

These errors may have different business consequences. The threshold was selected to support the objective of identifying a larger proportion of churn customers, but the final decision should still include business judgement, available resources, and the possible cost of unnecessary retention activity.

The dashboard was designed to communicate model behaviour clearly. Predicted probabilities, predicted labels, actual outcomes, threshold values, and SHAP contributions were presented separately so that users could distinguish between the model output and the observed customer outcome.

Care was also taken when describing the SHAP explanations. A positive SHAP contribution was described as increasing the model’s predicted churn probability, while a negative contribution was described as reducing it. The explanations were not presented as evidence that a feature directly caused a customer to churn.

Fairness was another relevant consideration because the dataset contained demographic and personal characteristics, including gender, SeniorCitizen, Partner, and Dependents. These variables may affect predictions in some other ways across the customer groups.

We need to know that the thesis did not complete a full fairness audit using specialised subgroup fairness measures. Therefore, we should not assume that the model will perform equally across every demographic group. Before use in a real organisation, the model should be tested for possible differences in error rates, prediction quality, and treatment across relevant customer groups.

The results are also limited by the use of one public sample dataset. The model should not be applied directly to customers from another telecommunications organisation without additional validation, updated data, and review of the local business context.

The Power BI dashboard was designed to support understanding rather than to hide uncertainty. Users were provided with probability values, explanation factors, and performance measures so that they could examine both the strengths and limitations of the prediction system.

These ethical considerations supported a more responsible use of the developed system. The final model and dashboard should be treated as analytical tools that assist human judgement rather than systems that replace human responsibility.

Table 17 Main ethical considerations in the study

Ethical area	Approach used in the thesis
Dataset privacy	Public sample data without direct customer names or contact details

Ethical area	Approach used in the thesis
Customer identification	customerID preserved only for linking analytical outputs
Participant confidentiality	Participant codes P01 to P10 used during analysis and reporting
Public reporting	Names and identifying details excluded from public materials
Human responsibility	Model used to support rather than replace human decisions
Prediction errors	False positives and false negatives considered during threshold selection
Transparency	Probabilities, labels, performance, and explanations presented separately
Causal interpretation	SHAP contributions not described as causes of customer churn
Fairness	Possible demographic differences acknowledged
Generalisation	Results limited to one public sample dataset

3.15 Chapter Summary

This chapter provided the complete methodology behind the research undertaken in creating and testing an interactive visual analytics system. Starting from the choice of IBM Telco Customer Churn data to the testing phases. The original data file provided had 7043 customers, and 11 customers have the value TotalCharges as missing, so these were removed. Therefore, a cleansed data set was formed with 7032 customers.

Exploratory data analysis was conducted to gain insight into the overall distribution of the target, to characterize customers, explore numerical and categorical variable data distributions, explore correlation and identify any potential outliers. A conclusion drawn from the analysis was that Churn customer data was the class which is in the minority and would have been influential on the evaluation of the selected models and selection of thresholds.

The cleansed data was cleansed ready for machine learning, customer ids, the target variable and features were extracted.

The Churn target was encoded using 0 for non churn and 1 for churn. The 19 model input features were transformed into 45 numerical features through standardisation, preservation of the binary variable, and one hot encoding of categorical variables.

The dataset was divided into training and testing data using an 80 percent and 20 percent stratified split. The training dataset contained 5,625 customers, while the testing dataset contained 1,407 customers. The preprocessing transformer was fitted using the training data only to prevent information from the testing data from entering the model development process.

We chose 5 models to evaluate this time, these were: Logistic Regression, Weighted Logistic Regression, Decision Tree, Random Forest and Gradient Boosting. Five fold stratified cross validation was used to compare the models and complete limited parameter tuning.

I used the Tuned Gradient Boosting Classifier as my final model. It performed well in probability ranking and yielded the stable validation score, and it also produced output which is compatible with tree based SHAP explanations. The tuning parameters I used were 100 estimators, learning rate is 0.05, max tree depth is 2, min number of samples per leaf is 30 and subsample is 0.80.

A classification threshold of 0.27 was selected using out of fold predictions from the training dataset. The purpose of lowering the threshold from 0.50 was to identify a greater proportion of actual churn customers. The independent testing dataset was not used during model selection or threshold selection.

SHAP TreeExplainer was used to explain the final customer predictions. SHAP values were produced for all 1,407 testing customers and all 45 processed features. These contributions were then grouped into the 19 original customer features to improve interpretation.

The prediction and SHAP outputs were imported into Microsoft Power BI Desktop. A six page interactive report was developed to present customer risk, global model behaviour, individual explanations, customer segment comparisons, and final model performance.

The developed system was evaluated through a comparison with a static explanation baseline. Ten participants completed tasks using both systems. The evaluation

measured task accuracy, completion time, understanding, trust, decision support, usability, perceived usefulness, and system preference.

There were considerations used , called Ethical, they were there throughout the process of research. These considerations covered customer and participant privacy, responsible use of predictions, fairness, transparency, interpretation of SHAP explanations, and the limitations of applying results from one public dataset to other business settings.

The methodology described in this chapter created a connected process from raw customer data to model predictions, SHAP explanations, an interactive Power BI dashboard, and user evaluation. The next chapter presents the machine learning results, final model performance, threshold outcomes, and SHAP explanation findings.

Chapter 4 Results and Findings

4.1 Introduction to the Results

The findings from the stages of machine learning, threshold analysis, SHAP, Power BI, and evaluation will be presented in this chapter. The findings are presented in the same general order as the research process described in Chapter 3.

The results begin with a comparison of the five classification models evaluated using fivefold stratified cross validation. The performance of each of the model will be analysed in the terms of their accuracy, precision, recall, F1 measure, ROC AUC, Precision Recall AUC, and the stability.

The model comparison was followed by limited parameter tuning for Logistic Regression and Gradient Boosting. The Tuned Gradient Boosting Classifier was selected as the final model because it produced the strongest probability ranking results while maintaining stable validation performance.

The final model was trained using all 5,625 training customers. It was then evaluated using the independent testing dataset containing 1,407 customers, including 374 churn customers and 1,033 non-churn customers. The testing data had not been used during model selection, parameter tuning, or threshold selection.

The default classification threshold of 0.50 was compared with the selected threshold of 0.27. Reducing the threshold the model predicted more of the real churn customers but incorrectly predicted more non churn customer as churn.

The final model using the selected threshold obtained accuracy 0.7306, precision 0.4959, recall 0.8075, F1 Score 0.6144. The final model's ROC AUC is 0.8383, Precision Recall AUC is 0.6520 and the Brier score is 0.1386. The confusion matrix has values of true negatives are 726, false positives are 307, false negatives are 72 and true positives are 302.

In the conclusion, with this combination we can see that the model predicted 302 true positives from the real churn customers of which there were 374 true positive. From here there is results of explanation using SHAP which can explain each customer feature influencing on customer to churn, firstly explain each feature importance among all customers. Then explain particular customer from features influence which will explain each feature probability to affect specific churn customers. There were 1407 testing customers and 45 number of features, for all these parameters SHAP value were calculated for those features in model and their contributions grouped again under the original customer features for simplification. Then the final result from Power BI reports are display. The six report pages show

customer risk, global feature importance, individual explanations, customer segment comparisons, and model performance through an interactive visual interface.

Finally, the results of the user evaluation are presented. The comparison examines whether the interactive Power BI dashboard supported better task performance, understanding, trust, decision support, and usability than the static explanation baseline.

Together, these findings show the predictive results of the final model, the effect of the selected threshold, the main factors influencing its predictions, and the practical value of displaying the explanations through an interactive dashboard.

4.2 Baseline Model Comparison

Five basic models for classification were evaluated against the training data using five fold stratified cross-validation. The models were Logistic Regression, Weighted Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting. The test set did not play a role in this analysis.

Each of the evaluated models performed the best on one or more of the evaluation metrics. Some predicted customers that were likely to churn with greater accuracy and precision, others predicted a higher proportion of actual churning customers.

The highest accuracy was achieved by Gradient Boosting at 0.8041, along with the highest precision 0.6719. It had the strongest probability ranking performance during cross validation with ROC AUC 0.8494 and Precision Recall AUC 0.6683, yet a recall of 0.5157 indicated that only just more than half of the churners were identified on the cross validation sets. Logistic regression performance was similar to Gradient Boosting at a recall of 0.5472 and was second highest at accuracy at 0.8025.

It had a precision and ROC AUC of 0.6538 and 0.8461 respectively along with a PR AUC of 0.6601; a well-rounded model however, a large portion of customers predicted to churn or not churn (missed opportunity at catching churners).

Weighted Logistic Regression outperformed all models for actual churn customer classification with a recall of 0.8027 and F1 of 0.6324. The weighted logistic regression reported a ROC AUC of 0.8459 and Precision Recall AUC of 0.6589. The accuracy for the model was 0.7520, with a precision value of 0.5218, meaning although more real customers with churn were caught, more non churn people were identified as churn.

Random Forest performed somewhat close to a balanced result. It performed an accuracy of 0.7835, precision of 0.5777, recall of 0.6903 and a F1-score of 0.6289. Its ROC AUC and its Precision Recall AUC was 0.8437 and 0.6539, respectively. The results seem acceptable, and there was bigger difference between training and validation performance.

Decision tree also found many churn customers from the actual ones (recall = 0.7967). Also, its F1 score was 0.6157 and its precision was 0.5038. The accuracy was 0.7358, ROC AUC was 0.8301 and Precision Recall AUC was 0.6054. This model can be used to distinguish churn customers but is not performing very well in ranking customers by their churn risk.

Overall, the results showed that improving churn detection caused in the reduction of accuracy or precision. Weighted Logistic Regression and Decision Tree found out a larger number of churn customers on the other hand Gradient Boosting produced the strongest ROC AUC and Precision Recall AUC results. As the dashboard was designed to show the churn probabilities and customer risk levels but also the ranking performance was important. For this reason, Gradient Boosting and Logistic Regression were taken in account for further parameter tuning. The complete comparison is shown below in the Table 18.

Table 18 Baseline model cross validation results

Model	Accuracy	Precision	Recall	F1 score	ROC AUC	Precision Recall AUC
Gradient Boosting	0.8041	0.6719	0.5157	0.5832	0.8494	0.6683
Logistic Regression	0.8025	0.6538	0.5472	0.5955	0.8461	0.6601
Weighted Logistic Regression	0.7520	0.5218	0.8027	0.6324	0.8459	0.6589
Random Forest	0.7835	0.5777	0.6903	0.6289	0.8437	0.6539
Decision Tree	0.7358	0.5038	0.7967	0.6157	0.8301	0.6054

4.3 Model Stability and Overfitting Results

Model performance on the training folds was compared with performance on the validation folds. This comparison was used to examine whether the models learned patterns that could also be applied to customer records that were not used during training.

A small difference between training and validation performance suggested that a model produced relatively stable results. A higher difference indicates how well the model performed on the training data but less on unseen validation data. Logistic Regression and Weighted Logistic Regression had rather small discrepancies between training and validation results, and showed similar performance across all folds of the cross validation which indicated low danger of large overfitting. Controlled Decision tree also showed a reasonably small difference between training and validation.

The maximum depth and minimum leaf size helped prevent the tree from becoming too complex and learning the training records too closely.

The baseline Gradient Boosting model produced stable validation performance. Its results showed that the sequential trees were able to learn useful customer patterns without creating a large difference between training and validation performance.

In the five baseline models, overfitting appears most strongly in the Random Forest model, which had an F1 score gap of approximately 0.1841 between training and validation scores. This represents that its training F1 score was significantly greater than its average validation F1 score. The Random Forest also produced ok results on the validation set. We do see the difference, meaning it appears to have fit the training data closer than the other two models.

This reduced confidence that the same level of performance would continue when the model was applied to new customer data.

The tuned Gradient Boosting model later achieved an F1 score difference of approximately 0.0165 between training and validation performance. This was much smaller than the gap observed for Random Forest and indicated that the tuned model remained stable.

The stable performance of Tuned Gradient Boosting was one reason it was selected as the final model. Its probability ranking results were strong, and the small training and validation difference suggested that these results were not produced only by fitting the training data closely.

The stability analysis was considered together with accuracy, recall, precision, F1 score, ROC AUC, and Precision Recall AUC. A model was not selected only because it performed strongly on its training data. It also needed to produce reliable results across the validation folds.

Table 19 Summary of model stability and overfitting results

Model	Stability observation	Interpretation
Logistic Regression	Small training and validation difference	Stable performance
Weighted Logistic Regression	Small training and validation difference	Stable performance
Decision Tree	Relatively small difference after complexity control	Limited overfitting
Gradient Boosting	Small training and validation difference	Stable performance
Random Forest	F1 score gap of approximately 0.1841	Higher overfitting risk
Tuned Gradient Boosting	F1 score gap of approximately 0.0165	Low overfitting risk

4.4 Final Model and Threshold Results

The Tuned Gradient Boosting Classifier was used as the final prediction model. Once the model was selected and tuned, this model was then evaluated on the independent testing data. The testing data set contained 1407 customers, with 374 actual churners and 1033 non-churners.

The model output a probability of churn for each of the test customers. I experimented with two classification thresholds for deciding whether to classify a test customer as 'Churn' or 'Non-churn'. The default threshold of 0.50, and the chosen threshold of 0.27.

If a customer predicted probability of churn is equal to or greater than the given threshold, it would be classified as 'Churn'.

If applying the default threshold of 0.50, the metrics were; Accuracy 0.7960, Precision 0.6495, Recall 0.5053, and F1 Score 0.5684. At this threshold the Accuracy and Precision are high; however, the recall only captured approximately 50% of actual churn customers.

If applying the chosen threshold of 0.27, the metrics were; Accuracy 0.7306, Precision 0.4959, Recall 0.8075 and F1 Score 0.6144. At this threshold the Accuracy and Precision drop but Recall is now very high. This threshold successfully captured approximately 80% of actual churn customers.

The ROC AUC and Precision Recall AUC were the same (0.8383 and 0.6520 respectively) across the two thresholds as well as the Brier Score (0.1386), as these are calculated using predicted probabilities not predictions after a set threshold.

From the thresholding analysis an important strategic trade-off can be seen. The 0.50 threshold captured few non-customers but missed out on almost 1/2 of the customers that actually churned. With the 0.27 threshold the churn customers are caught; however, at the expense of incorrectly identifying more non churn customers as churn.

A total of 185 customers who actually churned were missed if using the 0.50 threshold. Once the threshold was lowered to 0.27 this dropped to 72 customers. Therefore, by reducing the threshold by 0.23 this has reduced the number of missed churn customers by 61% or 113 customers.

The confusion matrix if using the 0.27 threshold contained 726 true negatives, 307 false positives, 72 false negatives and 302 true positives. Hence, for the 374 customers who actually churned, this model correctly identified 302 of them.

The 302 true positives (actual churn, predicted churn) could now be prioritised for customer retention activity. The 72 false negatives (actual churn, predicted non-churn) are also important, as this represents customers who are likely to leave and will not be offered retention efforts. The 307 false positives are customers who do not leave, but will be unnecessarily offered retention strategies. The 726 true negatives are non-churn customers who were correctly identified.

From an analytics perspective, the chosen threshold allows for a more sensitive customer risk analysis. For example, two customers might both have a 50% chance of churn with the 0.50 threshold, and in reality, both do not leave the company. With a threshold of 0.27, both could have a prediction of 'Churn' but a more in-depth analysis

might highlight that one of these customers has a 70% probability of churn, whereas the other customer has a probability of only 30%.

Instead, customers should be ranked based on predicted probability of churn. For instance, a customer with an 85% chance of churn, and another customer with a 30% chance of churn will, when evaluated at a threshold of 0.27 both be shown as 'Churn', but from an analytic and management perspective the customer with an 85% chance should be more of a priority for retention. By presenting both churn probability and predicted status, users can filter by probability, and compare across customers. The decision making as to which customers to target at for retention is then the responsibility of users.

The choice of 0.27 is not a universal recommendation for all companies. A company on a tight budget may choose a higher threshold to limit its customer outreach efforts, and another firm wanting to capture as much churn as possible, even at the cost of offering strategies to more non-churning customers, may choose an even lower threshold. The output of this model, in identifying a large number of actual churn customers provided the groundwork for its inclusion within the dashboard analytics.

Table 20 Final model results at the default and selected thresholds

Metric	Threshold 0.50	Threshold 0.27
Accuracy	0.7960	0.7306
Precision	0.6495	0.4959
Recall	0.5053	0.8075
F1 score	0.5684	0.6144
ROC AUC	0.8383	0.8383
Precision Recall AUC	0.6520	0.6520
Brier score	0.1386	0.1386

Figure 6 Comparison of final model performance at thresholds 0.50 and 0.27

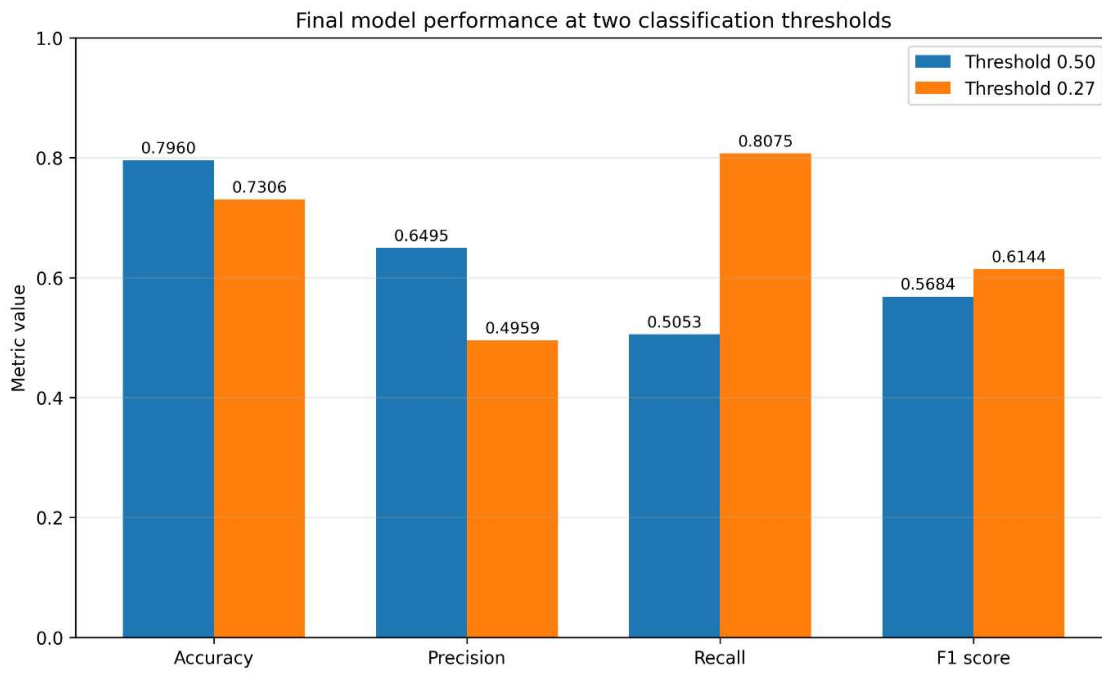
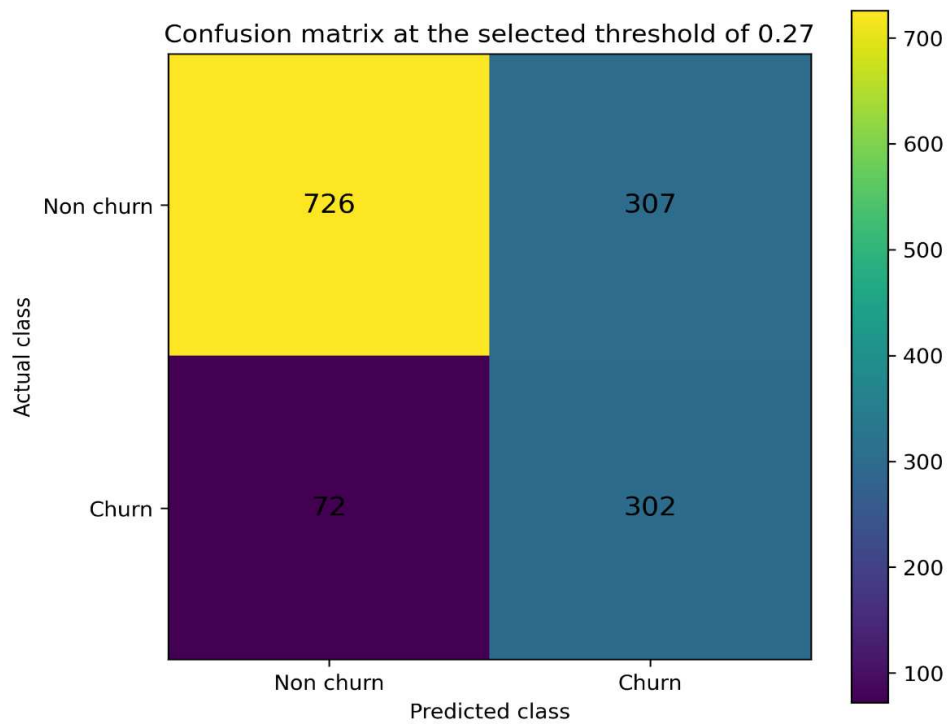


Figure 7 Confusion matrix of the final model at the selected threshold of 0.27



4.5 SHAP Analytics Results

SHAP explanations were generated to join the model predictions with the customer information that was shown in the Power BI dashboard. The purpose was not just to present the internal coding process but it was to help the users understand that which customer features had the greatest influence on the predicted churn probability.

SHAP values were calculated for all 1,407 customers in the independent testing dataset. Each customer initially received contribution values for the 45 processed model features. These processed values were then combined into the 19 original customer features so that the explanations could be presented using familiar names such as Tenure, Contract, Internet Service, and Monthly Charges.

A positive SHAP contribution indicated that a feature increased the model's predicted churn probability for a customer. A negative contribution indicated that the feature reduced the predicted probability. The contributions explain how the model reached its prediction, but they do not prove that the feature directly caused the customer to churn [2].

The global SHAP results showed that which of the features had the greatest total influence across the testing dataset. Where we can say that Tenure was the most important original feature later followed by Contract. Other important features included were Internet Service, Online Security, Technical Support, Payment Method, Monthly Charges, Multiple Lines, Paperless Billing, and Total Charges.

These results indicate that the model relied strongly on the length of the customer relationship, the type of contract, the services used by the customer, and the customer's payment and charge information. The importance values describe how strongly the model used each feature across all customers. They do not indicate that the same feature affected every customer in the same direction.

Global explanations are useful for obtaining an overall view of the model. For example, a business analyst can use the global importance ranking to identify which areas of customer information should receive greater attention during churn analysis.

However, global importance does not fully explain the prediction for one customer. Two customers with similar churn probabilities may have received those predictions for different reasons. Individual SHAP explanations were therefore created for every testing customer. For each customer, the explanation identified the strongest feature that increased churn probability and the strongest feature that reduced it. The remaining customer features were also ranked according to the strength and direction of their contribution.

This individual information was prepared together with the customer's predicted churn probability, predicted churn status, actual churn status, original feature values, and selected threshold. This made it possible to connect the prediction with a readable explanation rather than presenting only a churn label.

The customer explanations can also support the examination of correct and incorrect predictions. A user can investigate customers who were correctly identified as churn, customers incorrectly classified as churn, and actual churn customers missed by the model.

The grouped SHAP output contained 26,733 records, representing 1,407 customers and 19 original features for each customer. This format was used because it allowed Power BI to filter explanations by customer, rank contributions, compare features, and display positive and negative factors dynamically.

The SHAP results therefore provided the explanation layer of the analytics system. The churn probability showed the level of customer risk, while the SHAP values showed which customer characteristics influenced that risk according to the model.

From a business analytics perspective, this combination is more useful than presenting a prediction alone. Users can first identify customers with high predicted risk and then examine the factors connected to each prediction before making a retention decision.

The SHAP outputs were later presented through the Global Model Explanation and Individual Customer Explanation pages in Power BI. So, these pages allowed the users to move away from the overall model view towards a detailed explanation for a selected customer.

Table 21 Summary of the SHAP analytics outputs

Output	Result	Use in the analytics system
Testing customers explained	1,407	Customer level analysis
Processed model features	45	Detailed model explanation
Grouped original features	19	Business friendly explanation
Processed long format records	63,315	Detailed validation and analysis

Output	Result	Use in the analytics system
Grouped long format records	26,733	Main Power BI explanation table
Most important feature	Tenure	Global model analysis
Second most important feature	Contract	Global model analysis
Customer explanation outputs	Positive and negative contributions	Individual customer analysis

Figure 8 Global importance of the original customer features

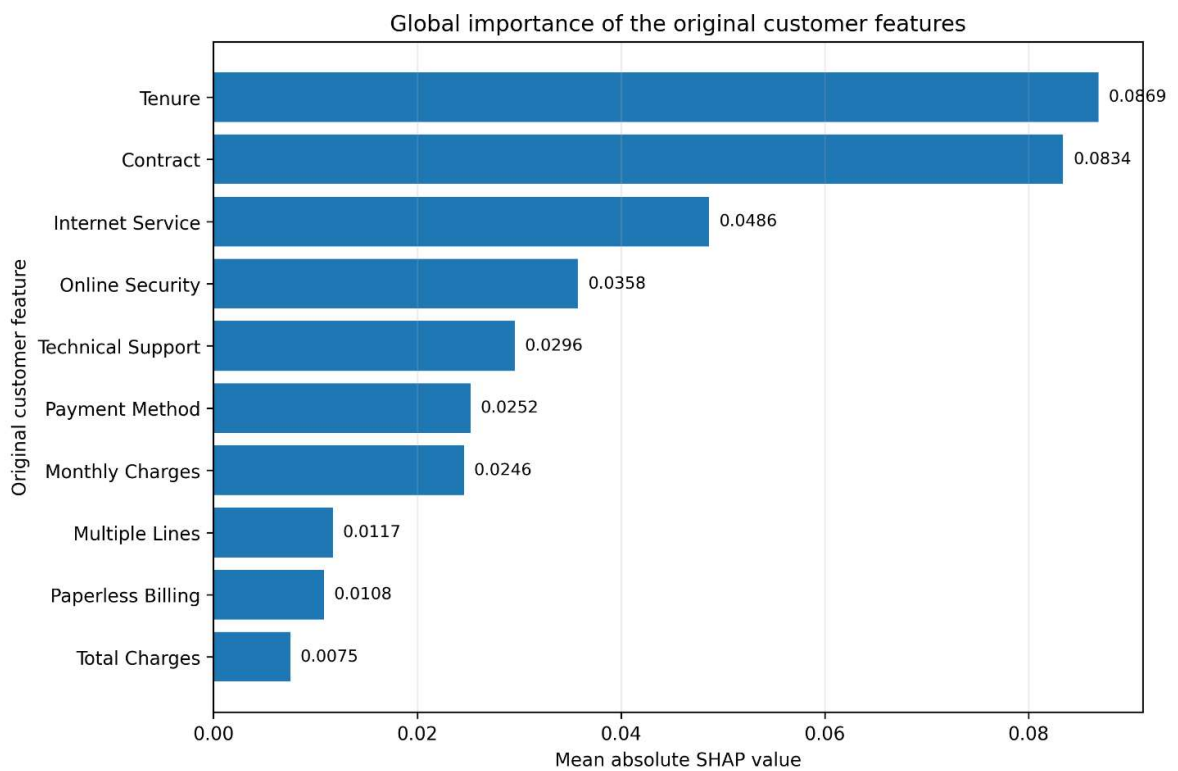
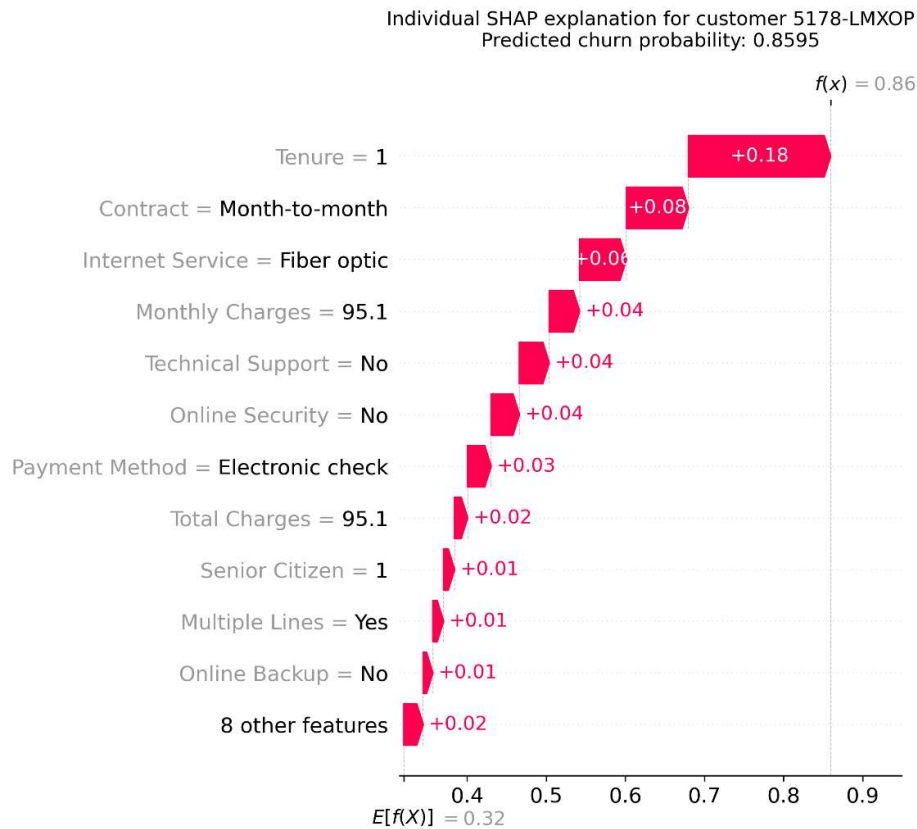


Figure 9 Example of an individual customer SHAP explanation



4.6 Power BI Analytics and Dashboard Results

4.6.1 Executive Overview

The Executive Overview page is a quick look at the prediction and explanation outcomes for the entirety of the testing dataset. This page is useful for understanding the overall customer pool you're working with before diving further into specific pages.

You can observe from the dashboard that the testing data set consisted of 1,407 customers. The data also showed that 374 customers did churn, and the model classified 609 customers as churn by use of the selected threshold of 0.27.

Based on these values, the predicted churn rate was 43.28% compared to an actual churn rate of 26.58%. The reason you saw a higher predicted churn rate is due to the lower prediction threshold, that you used in order to see more customers potentially at risk of churning.

The average churn prediction probability in the testing set was 27.2%, and the main final performance outcomes such as the recall (80.75%), ROC AUC (0.8383), and the F1 score (0.6144) are presented.

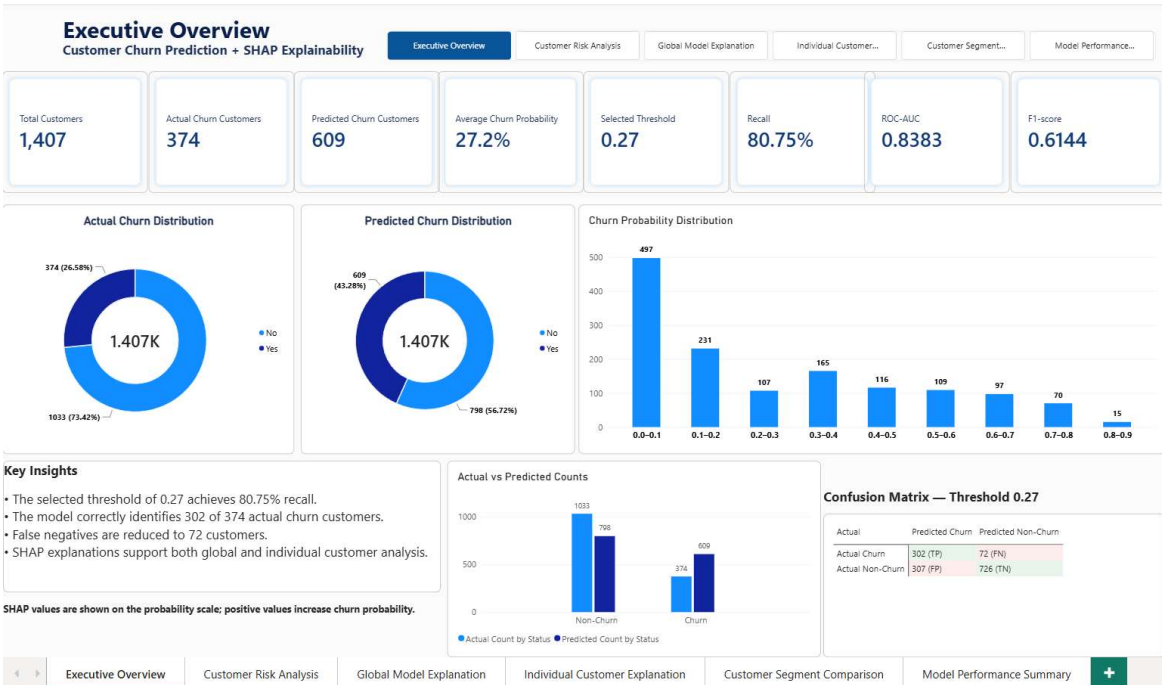
The churn probability distribution gives a more nuanced understanding than the prediction outcomes alone. For example, you'll see the group of largest predictions between 0.0 and 0.1 (497 total). The next largest category of risk, between 0.1 and 0.2, comprises 231 individuals. Conversely the highest risk categories contain 70 (between 0.7 and 0.8) customers, and 15 (between 0.8 and 0.9).

What is also evident from this distribution is that the level of risk was not solely confined to just 2 discrete groups. Instead, there are many levels of customer risk revealed by the probability values, suggesting more refined ways to triage your highest priority leads.

The confusion matrix, also available on this page, clearly illustrates the true predicted and actual outcome for the predicted customers. You'll also be able to find the numbers for the confusion matrix (which indicate 302 true positives, 726 true negatives, 307 false positives and 72 false negatives).

In essence, this page combines customer count, probability distribution, prediction outcomes, model performance, and explanation information all in one place, acting as your entryway into a more thorough dashboard analysis.

Figure 10 Executive Overview page of the Power BI analytics system



4.6.2 Customer Risk Analysis

The Customer Risk Analysis page was created to support more detailed investigation of customer churn risk. Users can filter the customer population using Contract, Internet Service, Payment Method, predicted churn status, actual churn status, and risk category.

The dashboard separated the 1,407 testing customers into 3 risk levels. The highest risk group contained 291 customers; the medium risk group contained 318 customers; the low risk group contained 798 customers. These groups represented 20.68% 22.60% and 56.72% of the testing population respectively.

The maximum predicted churn probability observed among the 1,407 customers was 85.95%. The customer closest to the selected threshold had a churn probability of 27.08%. When the customer closest to the threshold is showed it allows users to investigate boundary cases where even a small difference in the predicted probability may change the customer's predicted class.

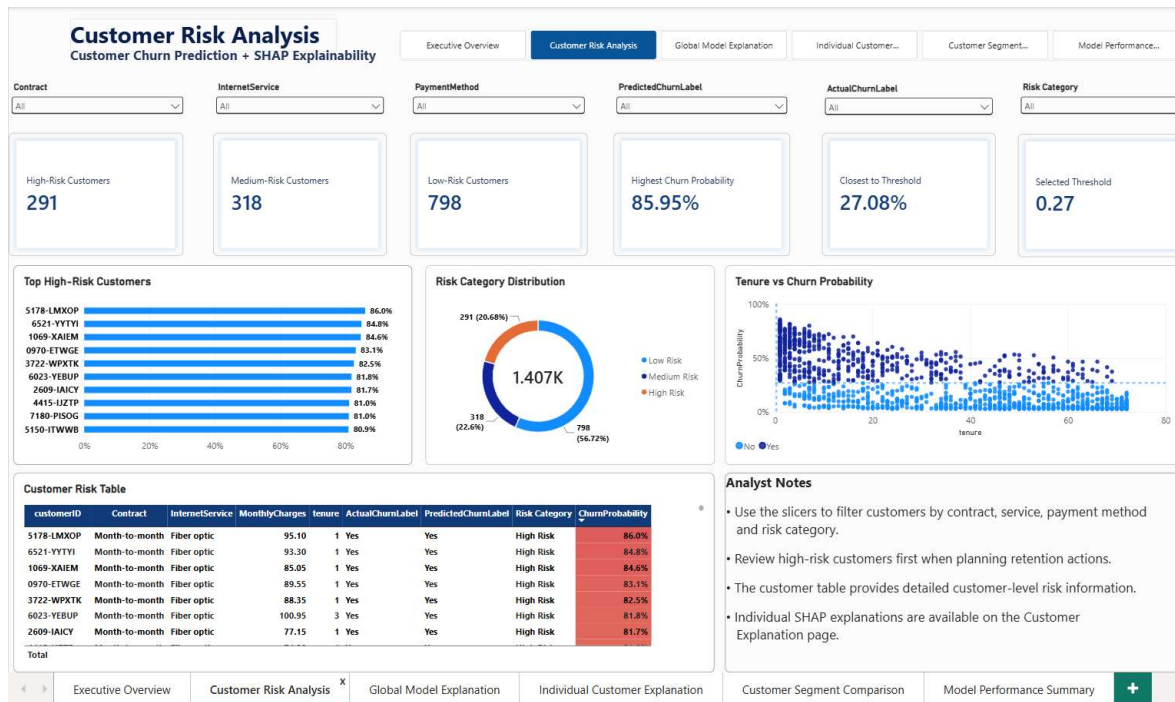
The top customer risk chart ranks customers according to their predicted churn probabilities. Customer 5178-LMXOP had the highest probability at approximately 86.0 percent. The other customers shown in the top risk group had probabilities between approximately 80.9 percent and 84.8 percent.

The relationship between tenure and churn probability is presented using a scatter chart. Customers with short tenure generally showed a wider range of churn probabilities, including many high risk values. Customers with longer tenure were concentrated more strongly in the lower probability area.

The page also includes a detailed customer table containing customerID, Contract, Internet Service, Monthly Charges, tenure, actual churn status, predicted churn status, risk category, and churn probability. This allows users to move from an overall risk view to detailed customer records.

From a business analytics perspective, this page supports the prioritisation of customers for further review. Users can begin with high-risk customers and then apply filters to identify groups that match a particular retention strategy or business condition.

Figure 11 Customer Risk Analysis page of the Power BI analytics system



4.6.3 Global Model Explanation

The Global Model Explanation page represents the overall factors that influenced the model across the complete testing dataset. This page displays around 19 original customer features instead of the 45 processed model features.

The explanation values are showed on the probability scale. The SHAP base value was 0.3203 this value showed the reference probability that was used when calculating the customer level contributions.

Here the Tenure was the most important global feature where it had a mean absolute SHAP value of approximately 0.0869. Contract had followed closely with an importance value that was around 0.0834. Internet Service ranked the third with the value around 0.0486.

Other important features involved were Online Security, Technical Support, Payment Method, Monthly Charges, Multiple Lines, Paperless Billing, and Total Charges. The importance chart helped the users to identify which customer characteristics were used very strongly by the model.

The direction chart provides additional information by showing the average signed contribution of each feature. However, an average direction should be interpreted

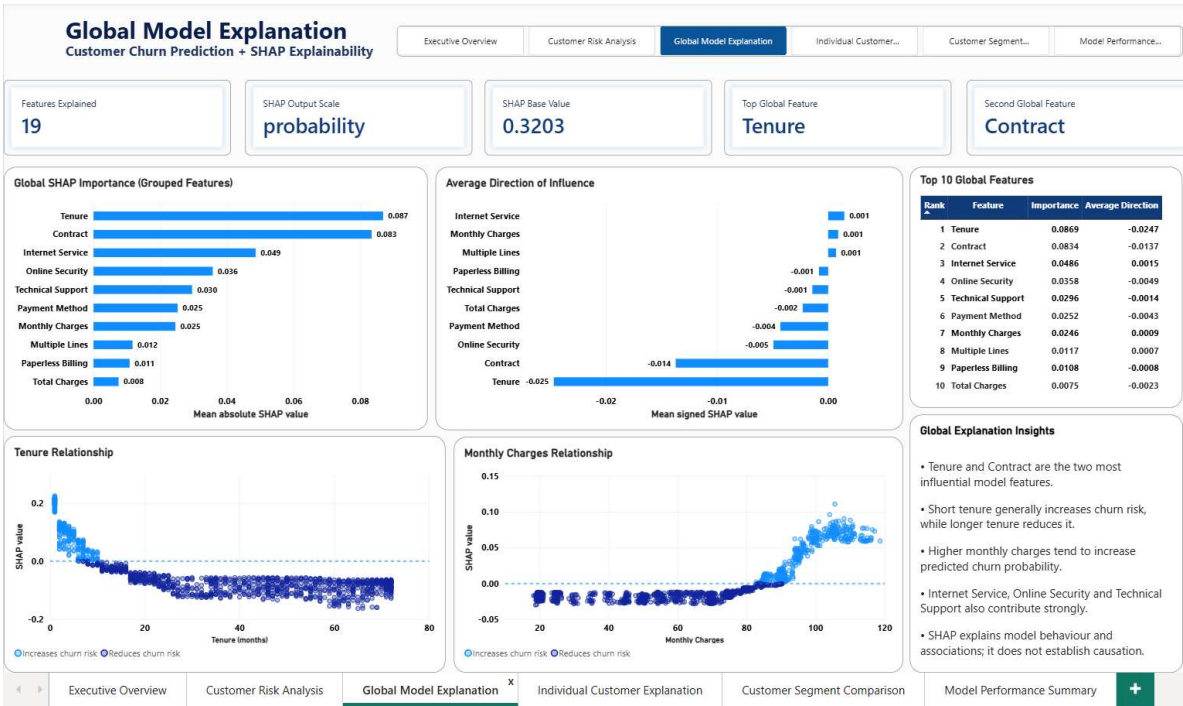
carefully because the same feature can increase risk for one customer and reduce it for another.

The tenure relationship chart shows that very short tenure was generally connected with positive SHAP values, meaning it increased the predicted churn probability. As tenure increased, the contributions became mostly negative, showing that longer customer relationships generally reduced the model's predicted risk.

The Monthly Charges chart represents that the lower and medium charges most generally produced smaller negative contributions. Higher charges that were mainly above approximately 80, increasingly produced positive contributions and increased the predicted churn probability.

This page provides analysts with a global view of model behaviour. It shows which customer areas are most relevant for further analysis without suggesting that the relationships prove direct causation.

Figure 12 Global Model Explanation page of the Power BI analytics system



4.6.4 Individual Customer Explanation

The Individual Customer Explanation page allows users to select one customer and examine the prediction in detail. The displayed cards and charts change according to the customer selected in the customerID slicer.

In figure 13 we can see the explanation for customer 5178-LMXOP. This customer had a predicted churn probability of 85.95 percent and thus was classified as churn. The actual result also showed that the customer really churned.

The customer's probability was considerably higher than the selected threshold of 0.27 and the base probability of 32.03 percent. The difference between the customer probability and the threshold was 58.95 percentage points.

The customer profile shows a month to month contract, tenure of one month, fibre optic internet service, Monthly Charges of 95.10, no Online Security, no Technical Support, and electronic check as the payment method.

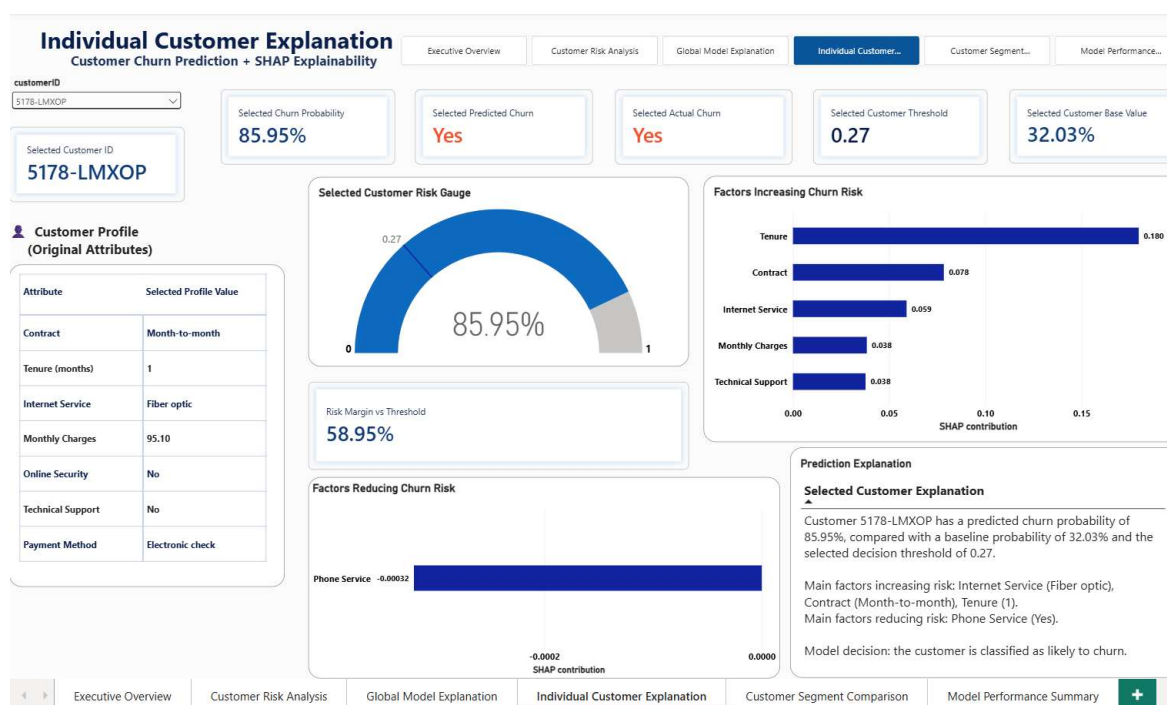
Tenure produced the strongest positive contribution which was with a SHAP value of approximately 0.180. The Contract contributed around 0.078, while Internet Service contributed around 0.059. Monthly Charges and Technical Support both contributed around 0.038.

Phone Service produced a very small negative contribution of approximately 0.00032. This contribution was seen to be too small to offset the stronger positive factors.

The page combines numerical information, customer attributes, visual contribution charts, and a written prediction explanation. This helps users understand why the model assigned a high probability rather than showing only the final churn label.

The same page can be used to investigate correctly predicted customers, false positives, false negatives, customers close to the threshold, and customers with very low predicted risk.

Figure 13 Individual Customer Explanation page for customer 5178-LMXOP



4.6.5 Customer Segment Comparison

On the “Customer Segment Comparison” page, you can explore different types of customers. The data on this page can be segmented by Contract, Internet Service, Payment Method, Senior Citizen status, and Paperless Billing. This comparison shows the notable differences between customer segments.

Average churn probability significantly differs from the type of contract customer has.

Customers with month to month contracts had the highest average churn probability at 42.5% when compared with 10.8% with one year contracts and 4.9% with two year contracts. In terms of actual churn by contract type the customers with the month to month contracts had the highest risk. This segment included 790 customers and had a predicted churn rate of 75.9%. The one year contract customers had a predicted churn rate of 3.1% and included 290 customers on the other hand the two year contract customers had a predicted churn rate of 0% and included 327 customers.

Internet Service type also demonstrated considerable differences.

FIBER OPTIC internet service had the highest average churn probability rate of 42.2% and a predicted churn rate of 70.1%. In contrast, the predicted churning rate with DSL internet service was 29.7%, and customers with no internet service had a predicted

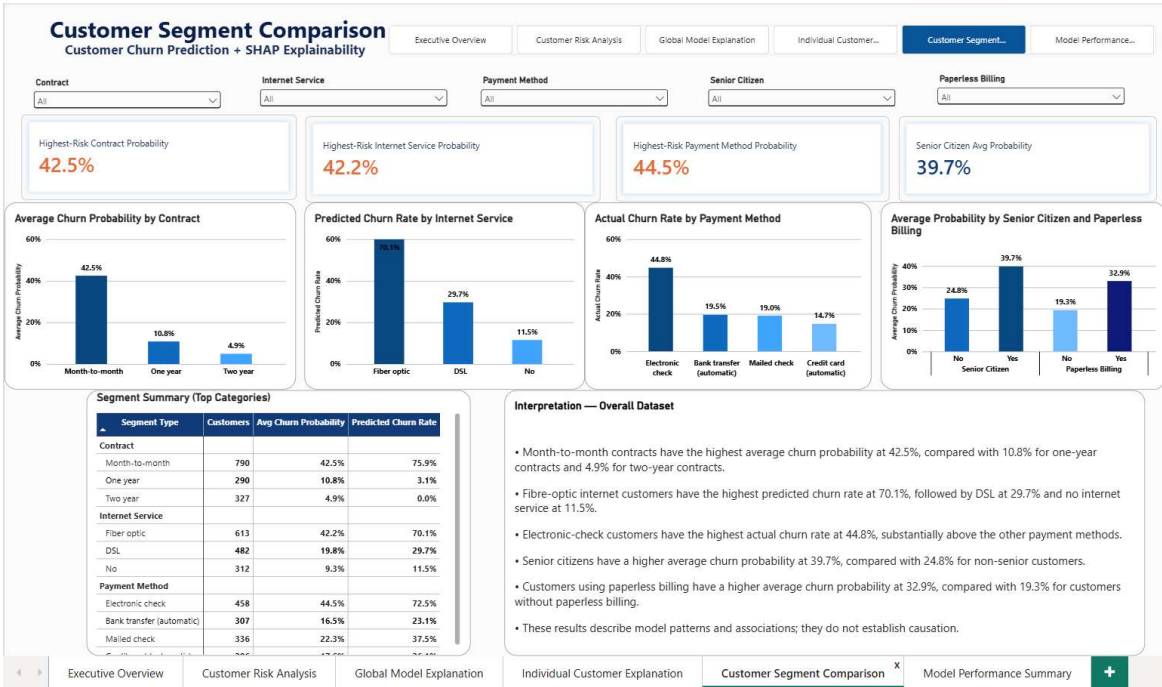
churn rate of 11.5%. The analysis on the actual churn based on payment type illustrated that customers paying with ELECTRONIC CHECK had the highest churn rate of 44.8%.

The actual churning rate for the remaining payment types was 19.5% for BANK TRANSFER, 19.0% for MAIL CHECK and 14.7% for CREDIT CARD. Electronic check customers also had the highest average actual churn probability at 44.5%. The analysis further indicates that the electronic check customer segment had a churn rate of 72.5% and consisted of 458 customers. Customers who were not senior citizens had an average churn probability of 24.8%, compared with 39.7% for senior citizens.

Customers who were using the paperless billing had an average predicted churn probability of 32.9%, compared with 19.3% for customers who were not using the paperless billing.

Each portion of the page demonstrates how the dashboard can be used to identify segments of customers with higher or lower potential and actual churn risks, such as those most likely to churn. These are excellent points where to begin deeper analysis or customer segmentation focused on loyalty building. However, keep in mind that the information derived herein describes the relationships found in the training and test data. Therefore, it should not be treated as cause-and-effect.

Figure 14 Customer Segment Comparison page of the Power BI analytics system



4.6.6 Model Performance Summary

The Model Performance Summary page outlines the final modelling results and the reasoning behind selecting a threshold of 0.27. Page includes accuracy, precision, recall, F1 score, ROC AUC, Precision Recall AUC, Brier score and selected threshold.

At threshold = 0.27 the final accuracy, precision, recall was all at 73.06%, 49.59% and 80.75% respectively. The F1, ROC AUC, PR AUC and Brier score are 0.6144, 0.8383, 0.6520 and 0.1386 respectively.

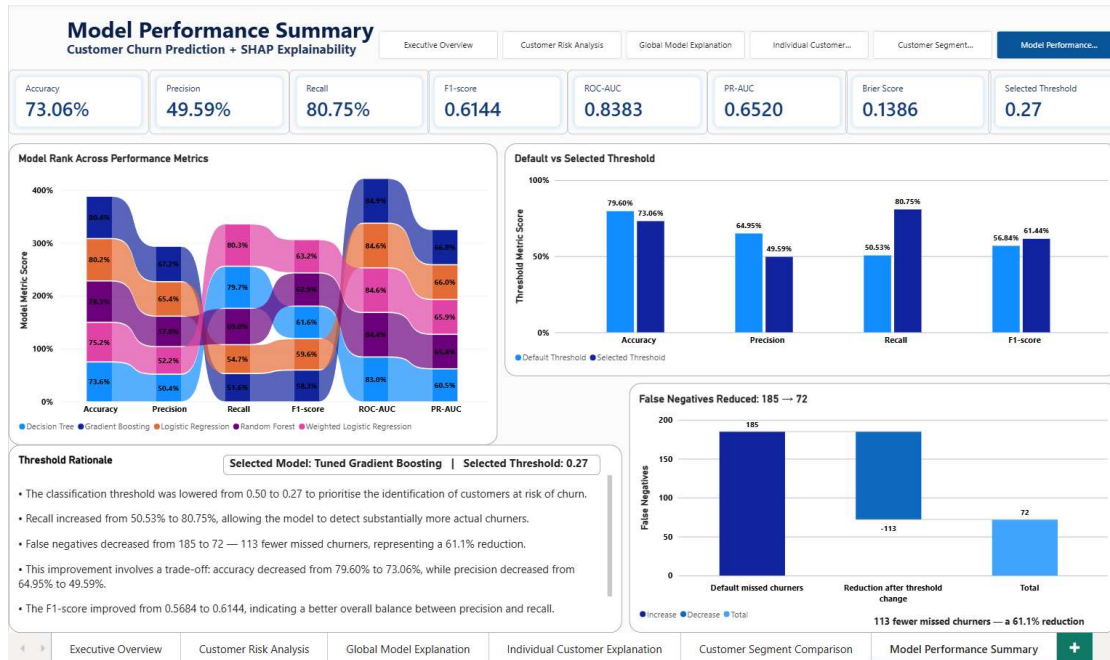
Chart showing threshold compared across major accuracy metrics illustrates the impact of lowering the threshold from 0.50 to 0.27. Accuracy decreases from 79.60% to 73.06%. Precision decreases from 64.95% to 49.59%. Recall however increases from 50.53% to 80.75%. F1 has been increases from 0.5684 to 0.6144. In summary, at threshold= 0.27 the models performed better balance for the goal of detecting churn customers who are actually churning.

The false negative chart shows that the number of actual churn customers missed by the model decreased from 185 to 72. This represents 113 fewer missed churn customers, or a reduction of 61.1%.

Finally, the Page plots compare the main performance metrics across the baseline models to show users that model performances differed according to metrics, the reason for not picking accuracy for the main performance metric.

The written threshold explanation makes the reasoning clearer by showing both the improved ability to identify churn customers and the increased cost of false positive predictions.

Figure 15 Model Performance Summary page of the Power BI analytics system



4.6.7 Overall Analytical Value of the Dashboard

The six dashboard pages form one connected visual analytics system. The report begins with a general overview and then allows users to move towards customer risk, global explanations, individual customer explanations, segment comparisons, and model performance.

The main analytical benefit is that users are not limited to one fixed result. They can filter customer groups, select individual customers, compare segments, examine risk probabilities, and review the factors connected with a prediction.

The dashboard also separates different levels of analysis. Global results describe the complete behaviour of the model, whereas the segment results compare the customer groups, and the individual results explain a single customer at a time.

The use of the probabilities and predicted labels supports more detailed decision making. Predicted labels provide a simple result, while probability values allow customers to be ranked and prioritised according to their level of risk.

The dashboard does not make automatic retention decisions on its own. But it organises the model results in a form that allows analysts and business users to review the evidence before selecting an action.

The completed Power BI report therefore represents the main analytics output of the thesis. It connects customer information, model predictions, SHAP explanations, performance results, and interactive analysis within one system.

4.7 User Evaluation Results

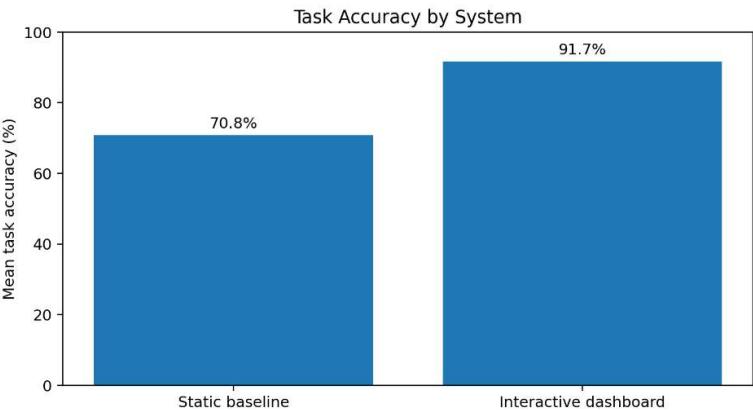
4.7.1 Task Accuracy

Tasks: Tasks accuracy was evaluated on the basis of six groups of tasks, representing a total of 12 points per system. The task groups span the “executive overview”, “customer risk”, “global explanations”, “individual customer explanations”, “segment comparisons”, and “model performance” domains. Static explanations the static explanation baseline obtained an average task score of 8.5 out of 12. This leads to a mean task accuracy of 70.83%.

Interactive Power BI dashboard This average task score was 11 out of 12 on the Power BI, which lead to a mean task accuracy of 91.67%. Discussion: The difference between the two conditions was about 20.83%. This indicates that users answered more questions correctly when they could freely browse among different pages, drill down to individual customers, filter their data and look at the information from a high level overview down to individual values. It turned out that the interactive Power BI dashboard was very useful to complete the tasks related to the individual customers, comparing different customer segments or fact checking on different pieces of information from more than one graphic element.

The static explanations included the correct information, but participants had to rely on the fixed screenshots that were provided, which made a detailed investigation more difficult.

Figure 16 Mean task accuracy for the static and interactive systems



4.7.2 Task Completion Time

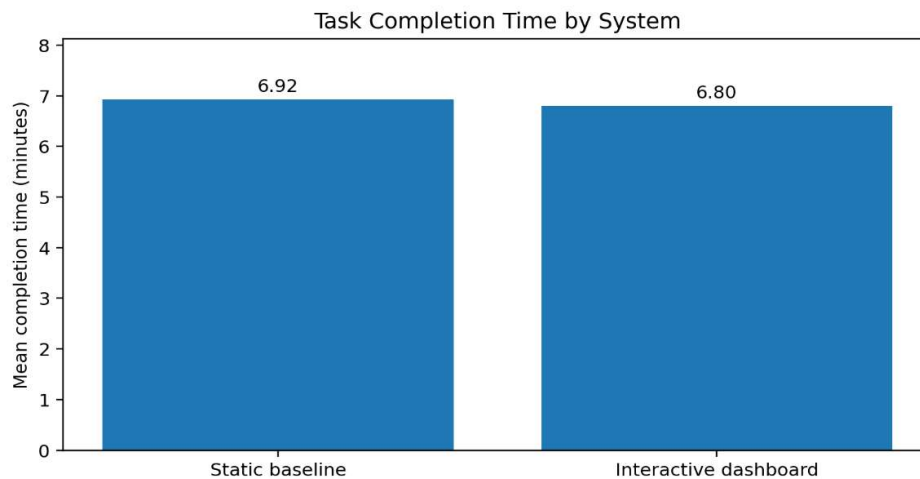
The mean completion time for the static baseline was mostly around 6 minutes and 56 seconds. The mean completion time for the interactive dashboard was mostly around 6 minutes and 48 seconds.

The interactive condition was therefore about eight seconds faster on average. The difference was small, but it also indicates that dashboard interaction did not create an overall time disadvantage.

Some participants without previous Power BI experience required more time to understand the page navigation and controls. But on the other hand, the other participants were able to locate the required information more quickly by using filters and moving directly to the relevant dashboard page.

The findings suggest that interactivity supported more correct and proper answers without increasing the overall completion time. The main improvement was therefore seen more in task accuracy rather than speed.

Figure 17 Mean task completion time for the static and interactive systems



4.7.3 Questionnaire Ratings

After completing their assigned tasks, the participants had to write feedback in a questionnaire that assessed both systems using ratings from 1 to 5. Measures included the understanding of system output, identification of churn reasons, decision support, trust, ease of use, and perceived usefulness. The average scores of both the systems as assessed by the participants were given in the table below.

The interactive dashboard performed very well in each dimension and got an overall average rating of 4.45 compared to 3.53 for the static explanations baseline, showing a proper 0.92 point improvement.

A key improvement was perceived by the participants in their understanding of the prediction and explanations (4.50 vs 3.50). By examining the details interactively, participants seemed to be better able to comprehend why the model was predicting that specific customer would churn. Participants' factor identification ratings also increased from 3.50 to 4.60.

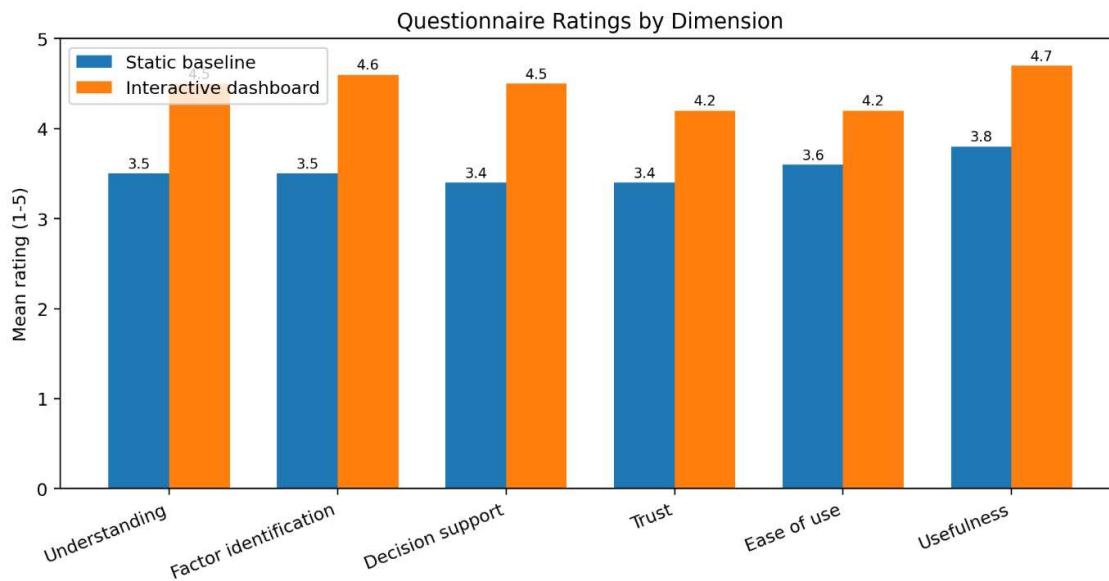
This higher rating tells us that the participants were better able to identify the individual factors behind a customer's predicted churn with the aid of dynamic customer selection and SHAP feature plots.

Finally, the participants also felt that the dashboard supported decision making on interventions to customers much more than static information (4.50 vs 3.40), the dashboard built trust in its decisions (4.20 vs 3.40), was easy to use (4.20 vs 3.60), and is far more useful for their churn/retention needs (4.70 vs 3.80). The increase in perceived usefulness for the dashboard (4.70) shows that not only are the individual aspects rated higher, but participants as a whole view this as a substantially more valuable product.

Table 22 Comparison of mean questionnaire ratings

Evaluation area	Static baseline	Interactive dashboard	Difference
Understanding	3.50	4.50	+1.00
Factor identification	3.50	4.60	+1.10
Decision support	3.40	4.50	+1.10
Trust	3.40	4.20	+0.80
Ease of use	3.60	4.20	+0.60
Perceived usefulness	3.80	4.70	+0.90
Overall mean	3.53	4.45	+0.92

Figure 18 Mean questionnaire ratings for the static and interactive systems



4.7.4 System Preference

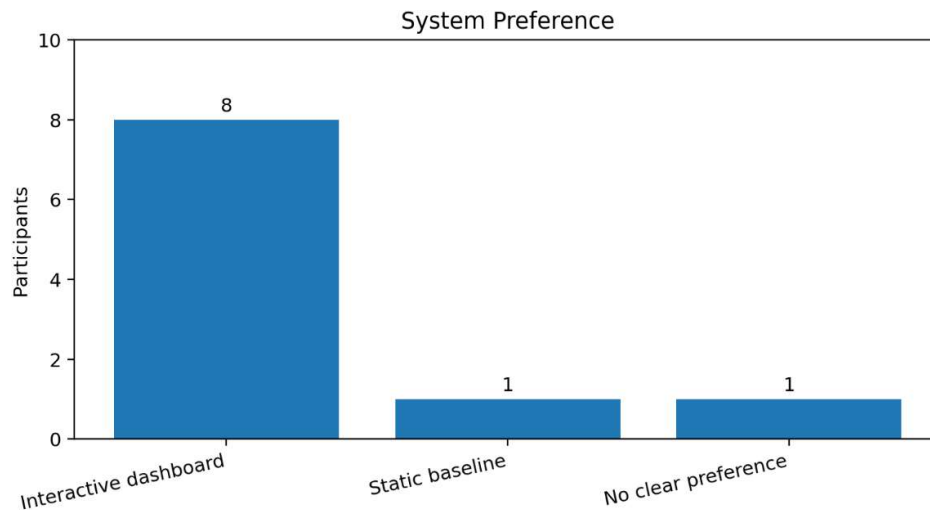
After evaluating both systems all the participants selected the version that they wanted. The 8 of the 10 participants preferred the interactive Power BI dashboard and one participant preferred the static explanation baseline, while the remaining one participant reported no clear preference.

The interactive dashboard was mainly preferred because it provided more detailed and on point information, allowed customer and segment comparisons, and made it easier to verify the reasons behind a prediction.

The static version was preferred by one participant because the information was immediately visible and no dashboard navigation was required. The participant with no clear preference recognised useful qualities in both systems.

The preference results support the task accuracy and questionnaire findings. Most participants considered the interactive system more suitable for detailed analysis, while the static version remained useful for a quick overview.

Figure 19 Participant preference between the static and interactive systems



4.7.5 Qualitative Feedback

The participant comments were reviewed to identify repeated positive points and limitations. The comments were used to provide context for the numerical findings rather than as a separate formal qualitative study.

The main strength of the static baseline was its simplicity. The values that were visible should be reviewed immediately, there was no need of learning any filters, slicers, or navigation controls.

The main limitations of the static version were that, the inability to change the selected customer, apply filters, compare alternative customer groups, or investigate cases that were not already displayed.

The interactive dashboard was valued for dynamic customer selection, easier comparison of customer segments, clearer examination of explanation factors, and stronger support for customer retention decisions.

The main limitation of the dashboard was that users needed a short period to become familiar with the six report pages and the available controls. This was more noticeable among participants without previous Power BI experience.

Overall, the feedback described the static baseline as suitable for a quick review and the interactive dashboard as more suitable for detailed investigation and decision support.

4.7.6 Overall Evaluation Finding

The overall evaluation results showed that the interactive Power BI dashboard performed and worked better than the static explanation baseline in most of the areas. It produced higher task accuracy, stronger questionnaire ratings, and a clear participant preference.

The interactive dashboard did not only improve subjective ratings. Participants also provided more correct task answers when using it. This is important because a dashboard that appears useful does not necessarily improve the accuracy of analysis.

The largest rating improvements were found in factor identification and decision support. These areas were directly supported by interactive features such as customer selection, filtering, connected pages, and dynamic visual updates.

The completion time was similar for both conditions. Therefore, the improved task accuracy was not achieved through a large increase in the time required to complete the tasks.

These results provide initial evidence that interactive visual analytics can improve how users understand and apply machine learning predictions and explanations. However, the participant group was small, so the findings should be treated as results from a practical initial evaluation rather than conclusions that can be generalised to all business users.

4.8 Chapter Summary

This chapter presented the final results, obtained for the machine learning model, threshold analysis, SHAP explanations, Power BI dashboard and user evaluation.

Firstly, five classification models were compared. Using 5 fold stratified cross validation. It was determined that Gradient Boosting demonstrated the strongest probability ranking but, Weighted Logistic Regression and Decision Tree performed better at identifying those in reality churners.

Stability analysis also highlighted Random Forest as having the highest risk of overfitting.

The Tuned Gradient Boosting Classifier was chosen as the final model. For the holdout test dataset, we found out that the accuracy was 0.7306, precision was 0.4959, recall was 0.8075, F1 score was 0.6144, ROC AUC was 0.8383, Precision Recall AUC was 0.6520 and, Brier score was 0.1386 with a selected threshold of 0.27. Lowering the

threshold from the default 0.5 to 0.27 increased recall from 50.53% to 80.75%. The number of missed churn customers decreased from 185 to 72, but this was done at the cost of incorrectly labelling more non churners as churners.

SHAP explanation on all 1407 customers in the test dataset was completed; it is noted that the 45 processed features were combined into 19 real customer features in order to allow easier comprehension of their implications.

The two global most important features are Tenure and Contract, while individual customers explained different attributes contributing to their respective increased and decreased churn predictions. The Power BI dashboard converted these machine learning and SHAP explanations to an interactive analytics system: it contains 6 pages for overview of the whole population; identify high-risk customers, global feature importance analysis; individual customer explanations, and comparison between customer segments and model performance comparison. Data from this dashboard showed that Month-to-month contract customers, Fiber optic customers, electronic cheque users, senior citizens and people with paperless billing have high predicted churn rates than the contrast group.

User study showed that the accuracy of the selected tasks performed using this interactive dashboard (91.67%, compared with 70.83% for the static system. The overall questionnaire rating was 4.45 for the interactive dashboard and 3.53 for the static baseline.) were much better, and eight out of ten participants preferred the interactive system over the static system (due to ability to choose any specific customers and compare differences among them). Overall, the results showed a system which provided predictive insight as well as providing analytical capability enabling the user to understand those insights effectively. The next chapter summarizes the meaning, relation to the research questions, value of the dashboard, and limitations of this study.

Chapter 5 Discussion

5.1 Introduction

This chapter discusses the meaning of the results presented in Chapter 4. The purpose is not to repeat all numerical findings, but to explain what they show about customer churn prediction, model explanation, interactive analytics, and decision support.

The discussion first considers the performance of the final model and the effect of lowering the classification threshold. Particular attention is given to the balance between identifying more churn customers and producing additional false positive predictions.

The SHAP findings are then discussed from an analytical perspective. The global explanations showed that which customer features were most influential across the testing dataset on the other hand the individual explanations showed that customers with same churn probabilities could receive those predictions for different reasons.

The Power BI dashboard is a main part of the discussion because it converted the prediction and explanation outputs into an interactive analytics system. The dashboard allowed users to examine customer risk, compare customer segments, investigate individual customers, and review the model results without using Python or machine learning code.

The user evaluation is also discussed to understand whether the interactive dashboard provided practical advantages over the static explanation baseline. The discussion considers task accuracy, completion time, questionnaire ratings, system preference, and participant feedback.

The findings are then connected directly with the four research questions. This helps determine whether the study achieved its aim of designing and evaluating an interactive visual analytics system for the explainable machine learning.

Finally, the limitations of the dataset, model, SHAP explanations, Power BI dashboard, and user evaluation are considered. These limitations are important because the results are based on one public dataset, one final model, and a small participant group.

5.2 Interpretation of the Final Prediction and Threshold Results

Final Model Insights With the final model outputs, it's evident that a model focused on prediction cannot be solely measured on accuracy alone. With the default threshold of 0.50, higher levels of Accuracy and Precision were seen, only around half of the customers who actually churned were identified by the model, making it less useful to an organisation that may wish to target its more at-risk customers.

After altering the threshold to 0.27 recall increased from 50.53% to 80.75% and while accuracy decreased to 73.06% (down from 79.60%), precision also decreased to 49.59% (down from 64.95%). The increased rate of identifying more potential churn customers, has come with a higher number of predictions in error.

This trade off was suitable for the main purpose of the study. The dashboard was designed to help users identify customers who may require retention attention. Missing an actual churn customer could mean that no action is taken before the customer leaves. A false positive may result in unnecessary retention communication, but the customer can still be reviewed before any expensive action is taken.

At the selected threshold, the model correctly identified 302 of the 374 actual churn customers. The number of missed churn customers decreased from 185 at the default threshold to 72 at the selected threshold. This represented 113 fewer missed cases, which was a reduction of approximately 61 percent.

However, the model also classified 307 non churn customers as churn. This means that the predicted label should not be treated as a final business decision. A customer classified as churn should first be examined using their probability, customer profile, segment information, and explanation factors.

The precision result of 49.59 percent shows that approximately half of the customers classified as churn actually churned. This may appear low when considered alone, but it reflects the decision to use a more sensitive threshold. The threshold was selected to increase the number of actual churn customers identified rather than to minimise every false positive.

The probability values are therefore more useful for analytics than the predicted class alone. Two of the customers have probabilities 0.30 and 0.85. Both the customers are at risk of churn for this selected threshold; however, not in same order of risk. We can prioritize the customer with probability 0.85 more.

This approach supports a ranked retention process. A company could first examine customers with the highest probabilities and then continue towards lower risk

customers depending on its available budget, staff, and retention capacity. The model therefore provides a risk ranking rather than a fixed instruction about which customers must receive an offer.

The selected threshold should not be treated as a universal recommendation. Different organisations may place different costs on false positives and false negatives. A company with limited retention resources may use a higher threshold and contact fewer customers. A company that considers missed churn very costly may accept a lower threshold and review a larger customer group.

The final ROC AUC of 0.8383 shows that the model had a useful ability to rank churn customers above non churn customers across different threshold values. The Precision Recall AUC of 0.6520 also showed that the model provided useful ranking performance for the smaller churn class. The Brier score of 0.1386 described the quality of the predicted probabilities.

These probability based results were especially relevant to the visual analytics system. Power BI presented the probability, risk category, predicted status, actual status, and explanation factors together. This allowed users to examine the uncertainty behind the prediction instead of relying only on a yes or no output.

Overall, the final model provided a practical basis for customer risk analysis. Its main value was not perfect classification. Its value was the ability to identify a large proportion of churn customers, rank customers according to estimated risk, and provide information that could be examined further through the interactive dashboard.

5.3 Interpretation of the SHAP Analytics Results

The SHAP results helped connect the model predictions with the customer information used in the dashboard. Instead of presenting only a churn probability or a predicted label, the system also showed which customer features influenced each prediction.

At the global level, Tenure and Contract were the two most important features. The implication here is that the length of the customer's relationship with the company and contract terms heavily influenced model's prediction of churn risk. The Customers with higher contract duration, were correlated more often than not to a low predicted risk, and shorter contract durations to a high predicted risk. Similarly, a monthly contract was found to have a high predicted churn, then those with year and two-year contracts.

Other significant predictors that affected risk of churn included Internet Service, Online Security, Technical Support, Payment Method, Monthly Charges, Multiple Lines, Paperless Billing, and Total Charges. This would indicate, that the prediction of customer churn was largely determined by aspects such as customer longevity with the company, use of products offered, payment characteristics and associated customer charges.

While the global outcomes were significant for establishing general trends across the whole testing dataset; these predictors would not be unique in their impact for each individual customer. A feature that would lead to increased risk profile for a particular customer may lead to minimal or even reversed impacts on another customer. This would be reason why providing individual explanations would have been invaluable. Thus the dashboard would provide which individual characteristics led to an increased predicted churn probability as well as which ones to decreased it on an individual customer basis.

For example, one customer may receive a high churn probability because of short tenure and a month to month contract. Another customer with a similar probability may be influenced more strongly by high Monthly Charges, fibre optic service, or the absence of Technical Support.

The individual explanations therefore made the predictions more useful for customer analysis. A user could examine the customer profile, churn probability, strongest positive factors, and strongest negative factors before deciding whether the customer required further attention.

The readability was made better by the grouped SHAP structure. The 45 processed model features were combined into 19 original customer features. This meant that users could work with familiar business variables rather than technical encoded feature names.

From an analytics perspective, SHAP was most useful when combined with the Power BI dashboard. The explanation values could be filtered, ranked, and displayed for different customers and segments. This allowed users to move between an overall model view and a detailed customer level view.

However, the explanations must be interpreted carefully. SHAP shows how the model used the available features to produce a prediction. It does not prove that a feature directly caused a customer to churn.

The SHAP results therefore supported transparency and interpretation, but they did not replace business judgement. The final decision about the retention action should

still consider customer history, available resources, business costs, and information that may not be there in the dataset.

So totally the SHAP analysis added value by turning the model output into information that could be explored and discussed. The main benefit was not the technical calculation itself, but was the ability to connect customer risk with understandable features in the dashboard.

5.4 Value of the Power BI Analytics Dashboard

The Power BI dashboard served as the primary analytics output of this study. It consolidated the customer and churn probability data along with the prediction, actual label, SHAP explanations, segment, and performance measures into a single, interactive report. A major benefit of this system was that business users could analyse this information without interacting directly with any Python code or the features and ML algorithms associated with it.

Instead, the results were presented in easily interpretable cards, charts, tables, slicers, and navigation buttons which anyone could interact with regardless of their technical background.

The six pages in the dashboard allowed the data to be viewed from several perspectives: The Executive Overview provided a summary of the population at risk and the final model performance. The Customer Risk Analysis viewed the high, medium, and low risk customers as sets to inspect in more detail. The Global Model Explanation summarized what, in the dataset as a whole, drove the predictions. The Individual Customer Explanation allowed a user to focus in on a single specific customer's prediction, SHAP explanation, and model score, while the Customer Segment Comparison analyzed groups of customers and finally the Model Performance Summary explained the final metrics as well as the benefits of lowering the classification threshold.

This page design was useful as it allowed the user to navigate from an overview of the data to individual predictions with a few clicks, identifying a group of at risk users, choosing one customer from that population, inspecting their churn probability and their contributing factors.

The filter and slicer tools allowed greater flexibility in exploring the dataset beyond just one fixed user population as users could filter according to contract type, internet service type, payment method, whether they are a senior citizen, whether or not they

receive paperless bills, the actual churn category, the predicted churn category, or a customer risk category. The segment data highlighted the business implications of the analytical data, for instance: it showed that those with month to month contracts had higher average churn risk than those with either a one year or two year contract. Both those with fibre optic internet and those who paid via an electronic cheque were more prone to churn than their comparisons on each of these parameters and those two sets would provide good target groups for further exploration by an analyst to further narrow down and explain them with model parameters.

Such analyses will enable business professionals to identify and more carefully research customer groups before considering lower-risk customers. This also supports focusing retention resources. Raising the threshold from 0.27 to 0.50 reduces the number of customers classified as churn from 609 to approximately 291. However, customers within the predicted churn group still have different probability values and should be prioritised according to their level of risk, but doesn't distinguish them into groups such that the highest risk ones can be examined more closely in one step.

Users could, however, sort the classified customers by probability and start reviewing them from the top down and therefore use retention resources efficiently in this regard.

Having the explanation accompany each prediction is critical. At its most basic, a high churn probability will not tell someone how to proceed with addressing customer churn. A high churn score on the Individual Explanation Page displays the prediction, SHAP graph as well as the contributing values so users know why the model decided that particular customer was high churn risk, for example short contract term, month-to-month billing, fibre optic, high monthly charge, and no Technical Support will all likely make a customer high churn risk. Transparency into whether the prediction appears correct from business and personal knowledge is crucial and this display allows that analysis.

Furthermore, showing the actual, predicted class, as well as the confidence for it displays what type of mistake is occurring: the false positives and false negatives.

Analysing false positives can help users identify customers who might receive unnecessary retention attention, and false negatives provide important information where and why errors occurred in predicting churn as an example where improvement may be necessary. The model performance chart displays the trade off of lowering the threshold so business stakeholders can decide where they fit between recall and precision: which business is better for and the trade offs incurred by their decisions, as

there isn't a single optimal threshold applicable to every business. This tool does not seek to eliminate human judgment; it intends to support a decision being made.

Final decisions would have business considerations about revenue vs loss, customers, available retention resources, among other factors not available in the modelling dataset. Evaluation showed that users of the interactive system were more accurate and rated their experience higher across a range of factors from understanding and ability to identify factors, to ease of use, trusting the system and ultimately how well the tool facilitated analysis and decision making during the analysis of customer churn. In conclusion the BI system has two primary benefits: organization of complex outputs, and interactivity to probe and manipulate these complex datasets.

The result is a visual environment which is useful as a whole throughout a churn analysis.

5.5 Interpretation of the User Evaluation Results

The user evaluation examined that whether the interactive Power BI dashboard supported understanding and decision making more effectively rather than the static explanation baseline. The two types of methods were compared descriptively across the evaluation measures.

Mean task accuracy improved by 20.83 percentage points when participants using the interactive dashboard rather than static display (static:70.83%, interactive: 91.67%). As far as the task goes, participants were able to find customer's risk score, explanation factors, and model information more correctly by the dashboard and controls. The ability to filter customer and compare segment by clicking on dashboard probably caused improvement, since participants of static method only see provided information.

The completion time did not change much: static system: around 6 minutes and 56 seconds; interactive dashboard: 6 minutes and 48 seconds. This result is relevant since interaction system often tends to need extra time to learn its navigation, controls and etc. But the additional controls did not slow participants in our experiment. Instead, participants finished tasks with greater accuracy and around the same time with using static method.

Overall mean score in the questionnaire rating indicated that interactive method provided higher scores: static: 3.53; interactive: 4.45. The areas of largest increase were factor identification (1.10) and decision support (1.10), revealing that interactive

dashboard could deliver better explanations with clearer factors to assist decision making.

The understanding score also gained 1.00 increase from static to interactive dashboard (static:3.50, interactive:4.50). This indicates that participants found customer churn information easier to understand with all information of prediction, individual customer details and explanation factors being provided.

The trust score increased from 3.40 to 4.20, which may attribute to that individual explanation and model information could build confidence for participants judging the predictions; nevertheless, it measures perceived trust rather than the objective trustworthiness of the system of the system.

Easier to use with smaller gap (static:3.60, interactive:4.20) between two methods probably is because most of participants are not familiar with Power BI functions; so, it could take some time to learn how to use filters and slicers. Participants with greater Power BI experience might have rated the ease of use more highly because they were already familiar with filters and slicers.

Highest score with interactive method is perceived usefulness (4.70). It confirms that participant find its practical usage for combining all information together to form a prediction, explanation, and analysis report within a single tool.

Another finding in preferring the system supported that eight out of ten participants prefer interactive system while one static, one no difference. Participant's preferences mainly attributed the interactive dashboard better for searching particular customers and performing analysis and comparative tasks such as filtering and zooming. Still static reports were beneficial for quick reference because it provides the most salient information without requirement of interactive exploration. Therefore, based on the result, static system is suitable for quickly providing information or communicating a particular fact efficiently whereas interactive dashboards facilitate exploration over certain elements among customers and model.

All results from this experiment should be considered with caution since there were only 10 participants with heterogeneous experience of Power BI and analytic expertise. The tasks used in this experiment were not as natural as in the real world for conducting customer retention business, and the tests were conducted in a controlled experimental environment. Statistical significance tests were not performed accordingly, all improvement represents the observation with this set of participants and we could not confirm its overall consistency for every user and organization.

The results do not imply static explanations are less valuable as they are easy to disseminate across different viewers. Static analysis can indeed communicate the salient aspects of an issue, which is very critical in a busy corporate environment. Interactive dashboards are generally preferable for analysis of individual records, exploration over different segments and investigations for individual reasons behind predictions.

Despite these limitations, the findings provided the initial practical evidence that the Power BI dashboard supported more accurate analytical decision making with stronger perceived understanding, and better decision support than the static explanation baseline.

5.6 Answers to the Research Questions

5.6.1 Research Question 1

How effectively can machine learning predict customer churn in the selected dataset?

The results show that machine learning could identify useful churn patterns in the selected telecommunications dataset. It had a good balance of ranking performance, stability and generalisation, and was ultimately chosen out of the five models evaluated.

When set to a threshold of 0.27, our chosen model successfully attained an accuracy of 73.06% (with a true positive of 302/374 churned customers), a recall of 80.75%, F1 score of 0.6144, and ROC AUC of 0.8383. The trade-off between reducing the threshold, identifying more churn customers, and producing more false positive predictions can be observed. Setting the threshold of our final model to the default of 0.50 results in greater accuracy and precision, but allows 185 of the churners to go undetected. Lowering it to 0.27, successfully reduced the missed churners to 72.

Therefore, the model was effective as a customer risk identification and ranking tool. However, it did not produce perfect predictions. Its outputs should be used to support customer review and retention planning rather than as automatic final decisions.

5.6.2 Research Question 2

Which customer features have the strongest influence on churn predictions at the global and individual levels?

The global SHAP results showed that Tenure and Contract were the two most influential features in the final model. Internet Service was the third most important feature.

Other important variables that included were the Online Security, Technical Support, Payment Method, Monthly Charges, Multiple Lines, Paperless Billing, and Total Charges. These results indicate that the churned risk was influenced by the customer relationship length, contract conditions, service selection, payment behaviour, and charges.

The segment analysis supported these findings. Customers with month to month contracts showed higher average churn risk than customers with one year or two year contracts. Fibre optic customers and electronic cheque users also showed higher churn risk than their comparison groups.

At the individual level, the important factors varied between customers. Short tenure and a month to month contract strongly increased the prediction for some customers, while Internet Service, Monthly Charges, Technical Support, and Payment Method had stronger effects for others.

Therefore, global explanations identified the main patterns used across the complete testing dataset, while individual explanations showed the specific combination of factors connected with each customer prediction.

5.6.3 Research Question 3

How can machine learning predictions and explanations be presented in Power BI so that nontechnical users can understand and explore them?

The study presented model predictions and SHAP explanations through a six page interactive Power BI dashboard. The system combined customer profiles, churn probabilities, risk categories, predicted outcomes, actual outcomes, global feature importance, individual explanation factors, segment comparisons, and model performance.

Cards, charts, tables, slicers, and page navigation were used instead of technical code or processed feature outputs. The 45 processed features were grouped into 19 original customer variables so that users could view familiar business terms.

The dashboard allowed users to begin with an overview of the testing population and then move towards segment or individual customer analysis. Users could filter customers, select individual records, compare groups, and inspect the reasons connected with a prediction.

The user evaluation indicated that participants generally understood the information more effectively when using the interactive dashboard. The understanding rating increased from 3.50 with the static baseline to 4.50 with the interactive system.

Therefore, predictions and explanations can be made more accessible by organising them into different analytical levels and allowing users to explore them through familiar visual controls.

5.6.4 Research Question 4

Does an interactive visual analytics system support user understanding and decision making more effectively than a static explanation?

The evaluation results mostly supported the interactive dashboard. Mean task accuracy increased from 70.83 percent and then the static explanation was increased to 91.67 percent with the interactive system. This showed an improvement of 20.83 percentage points.

The overall questionnaire rating increased from 3.53 to 4.45 out of 5. The largest improvements were found in the factor identification and the decision support, with these two measures increasing by 1.10 points.

Completion time remained similar. Participants required approximately 6 minutes and 56 seconds with the static system and 6 minutes and 48 seconds with the interactive dashboard. This suggests that the additional interactive functions did not create a clear time disadvantage during the evaluation.

Eight of the 10 participants preferred the interactive system. Their feedback mainly referred to the ability to select customers, apply filters, compare segments, and examine explanation factors.

The 10 participants help to achieve these findings, and no statistical significance test was performed. The results therefore provide initial evidence for this study rather than proof that the interactive dashboard will always perform better in every business setting.

Overall, the interactive visual analytics system supported higher task accuracy, stronger perceived understanding, and better decision support than the static explanation baseline for the participants included in this evaluation.

5.7 Limitations of the Study

Although the developed system produced useful results, several limitations should be considered when interpreting the findings.

The first limitation concerns the dataset. The study used one publicly available telecommunications churn dataset containing 7,032 cleaned customer records. The results therefore represent the patterns available in this dataset and may not reflect the behaviour of customers in other organisations, countries, industries, or time periods.

The dataset also contained a limited set of customer variables. Important factors such as customer complaints, service quality, competitor offers, customer satisfaction, contact history, retention offers, and reasons for cancellation were not available. Including these variables could improve prediction performance and the usefulness of the explanations. The second limitation concerns the machine learning models. Five classification models were evaluated, but the study did not examine every possible algorithm or tuning combination. More advanced models might produce different results, although they could also increase complexity and reduce interpretability.

The selected threshold of 0.27 was suitable for the purpose of identifying a larger proportion of churn customers in this study. However, it was not selected using real financial costs. The actual value of a threshold should depend on the cost of a missed churn customer, the cost of unnecessary retention action, available staff, and the organisation's retention budget.

The model was evaluated using one train and test division with stratified sampling. Although cross validation was used during model comparison and threshold selection, evaluation on additional datasets or data collected during a later period would provide stronger evidence of generalisation.

The SHAP explanations also have limitations. SHAP shows how the model used the available variables when producing a prediction, but it does not prove that those variables caused the customer to churn. A high SHAP contribution should therefore be understood as model influence rather than direct causal evidence.

The grouped SHAP outputs improved the overall readability by combining 45 processed features into 19 original variables. But this grouping may also hide differences between individual encoded categories. The simple made presentation was useful for business users, but on other hand it provided less technical detail than the completely processed feature results.

The Power BI dashboard was developed as a desktop based analytical prototype and was not connected to a live organisational customer database. As such, aspects like automated data updates, scoring in real time, security and user permission control, cloud-based hosting and deployment, security issues, or customer relationship management systems integration were not tested in this study.

The dashboard also contains just one dataset and one churn prediction model. For a new business with different variables, customer classifications, business regulations, and reporting preferences, significant development work would be required.

Since the study was limited to 10 individuals taking the usability test, one must bear in mind that sample sizes this small restrict any findings to the specific participants of the study. In addition, each individual participating in the study possessed unique skill sets as far as education level and Power BI and analytics experience.

Tasks performed were on a limited range of activities pertinent to customer churn analysis performed under tight controls and not necessarily in business circumstances. They did not fully reproduce the time pressure, organisational responsibilities, financial consequences, and collaboration involved in real customer retention work.

The study used descriptive comparisons between the static and interactive systems. No statistical significance test was performed. Therefore, the observed differences in accuracy, ratings, and completion time describe the performance of this participant group but do not confirm that the same differences would occur for all users.

The static baseline was also more limited than the interactive dashboard. It presented fixed information and did not allow customer selection, filtering, or segment exploration. This supported a clear comparison between static and interactive presentation, but some of the improvement may have resulted from the greater amount of functionality available in the dashboard.

Finally, the study measured short term task performance and participant opinions. It did not examine long term use, learning effects, continued trust, actual retention decisions, or financial outcomes. A longer evaluation within a real organisation would be needed to determine whether the system improves business performance over time.

These limitations do not remove the value of the study, but they define the conditions under which the results should be interpreted. The developed system should be viewed as a practical prototype and an initial evaluation of how machine learning, SHAP, and Power BI can be combined for explainable customer churn analysis.

5.8 Chapter Summary

This chapter shows us the findings from Chapter 4 and connects them back to the objectives of the study.

In conclusion, the final Tuned Gradient Boosting Classifier provided valuable insights for predicting churn. Setting the threshold to the selected value of 0.27 generated a recall of 80.75% correctly identifying 302 out of the 374 churned customers. By decreasing the threshold, the number of missed churning customers was lowered at the cost of a higher number of false positive predictions.

As such the discussion confirmed that the predicted outcomes should be viewed as an indication of customer churn risk rather than business decisions being automatically processed.

Predicted probabilities, individual customer profiles, segment analysis, and explanation features need to be jointly analyzed before intervention strategies are determined. The SHAP analysis was used to enhance the model's interpretability through the visualization of contributing features with predictions and it confirmed Tenure and Contract were the two most important global features, whilst, explaining contributions varied for customers based on different contributing factors. Finally, the interactive Power BI dashboard transformed the machine learning model and the SHAP explanation outputs to a visual analytics platform for which business stakeholders without prior machine learning expertise or Python programming knowledge can review customer risk, select specific customers, compare segments, inspect explanation drivers, and monitor model performance. The user evaluation research confirmed this with participants achieving higher task accuracy and giving higher ratings for understanding, ease of use, decision support, and usefulness when compared with the flat explanations baseline.

As predicted, a majority (8 out of 10) of the participants showed a preference for using the interactive system.

All of the research questions were successfully answered based on the outputs derived from the use of the machine learning models, SHAP, and the Power BI interactive dashboard, combined with an analysis of the results obtained from the user evaluation research. It was demonstrated how this combination of technologies, specifically a machine learning model, SHAP (a machine learning model, SHAP as an explainable artificial intelligence method, and Power BI as an interactive visual analytics tool), and an interactive visual analytics tool (Power BI), can successfully support customer churn analysis in organizations by providing not only predictions but also supporting tools

that promote a deeper understanding and exploration of those predictions amongst diverse user groups. This study was, however, limited by the use of one public customer dataset, the limited variety of customer features, the selection of one final machine learning model and one explanation method, the local development of the Power BI dashboard as an interactive prototype, the small user evaluation group of 10 participants, and the absence of statistical significance tests.

This chapter concludes by summarizing the findings and briefly outlines the final conclusions, the scientific contributions of this study, the practical implications and future research directions.

Chapter 6 Conclusion

6.1 Introduction

This chapter presents the final conclusions of the study. It summarises the main findings, explains the scientific and practical contributions, discusses the wider value of the developed system, and identifies possible directions for future research.

The study aimed to design and evaluate an interactive visual analytics system that combines machine learning, explainable artificial intelligence, and Power BI for customer churn analysis. The developed system included a churn prediction model, SHAP based global and individual explanations, a six page Power BI dashboard, and a user evaluation comparing the interactive system with a static explanation baseline.

The final machine learning model provided useful customer churn predictions and identified a large proportion of the customers who actually churned. The SHAP explanations improved transparency by showing the customer features associated with the predictions. Power BI converted these outputs into an interactive format that allowed users to explore customer risk, compare customer segments, and examine individual predictions.

The user evaluation provided initial evidence that the interactive dashboard supported higher task accuracy and stronger perceived understanding than the static baseline. Participants also gave higher ratings for factor identification, decision support, trust, ease of use, and perceived usefulness.

The conclusion does not repeat all detailed results from the earlier chapters. Instead, it brings together the overall outcome of the research and explains what the study contributes to the areas of machine learning, explainable artificial intelligence, and visual analytics.

6.2 Summary of the Main Findings

An interactive visual analytics system for understandable customer churn prediction was devised within the study. The final system utilized data preprocessing, machine learning modelling, SHAP explanations, Power BI visualisation, and user evaluation.

Five classification models were compared at the modelling stage. The Tuned Gradient Boosting Classifier was chosen as the final model due to its satisfactory ranking performance and generalization stability, suitable for the task of predicting churn risk of customers.

At the selected threshold of 0.27, the final model has achieved an accuracy of 73.06 percent and a precision of 49.59 percent and a recall of 80.75 percent and F1 score of 0.6144, and ROC AUC of 0.8383. From the 374 customers who actually churned, 302 were correctly identified.

By Decreasing the threshold from 0.50 to 0.27 reduced the number of missed churn customers from 185 to 72 and therefore making the model more useful for retention analysis. But the thing is that it also increases the number of false positive predictions compared with the higher threshold.

The SHAP analysis provided both global and individual explanations. At the global level, Tenure and Contract were the two most important features, followed by Internet Service and other customer service, payment, and charge related variables.

At the individual level, the explanations showed that customers with similar churn probabilities could receive those predictions for different reasons. This supported a more detailed understanding of each customer rather than relying only on a general churn label.

The 45 processed features used by the model were grouped into 19 original customer variables. This made the explanations easier to understand and more suitable for presentation in Power BI.

The six page Power BI dashboard combined customer profiles, predicted probabilities, risk groups, actual outcomes, SHAP explanations, segment comparisons, and model performance. It allowed users to move from an overall view of the customer population to detailed analysis of individual customers.

The dashboard analysis showed that customers with month to month contracts, fibre optic internet, electronic cheque payment, senior citizen status, and paperless billing generally had higher predicted churn risk than their comparison groups.

The user evaluation compared the interactive dashboard with a static explanation baseline. Mean task accuracy showed a good rise from 70.83 percent with the static system to 91.67 percent with the interactive dashboard.

The total question rating increased from 3.53 to 4.45 out of 5. The participants also gave more ratings for understanding, factor identification, decision support, trust, ease of use, and perceived usefulness.

Eight of the 10 participants preferred the interactive dashboard. The most valued functions were customer selection, filtering, segment comparison, and the ability to investigate explanation factors for individual predictions.

The time required to evaluate the two systems did not differ noticeably. This suggests that the extra interactivity enabled improved accuracy of the analysis without any noticeable performance cost at the evaluation stage. In summary, the research demonstrated that coupling machine learning, SHAP, and Power BI formed a system where a user was not only provided with predictive information, but also analytical tools for assessing factors contributing to a risk-oriented prediction and exploring these factors at the population, segment, and individual customer levels.

6.3 Scientific and Technical Contributions

This study makes several scientific and technical contributions to the use of explainable machine learning and visual analytics for customer churn analysis.

The first contribution is the development of a complete analytical workflow that connects data preprocessing, machine learning prediction, SHAP explanation, Power BI visualisation, and user evaluation. These components were not treated as separate tasks but were combined into one system through which model outputs could be explored by users.

The second contribution is the use of a classification threshold that reflects the purpose of customer retention analysis. A threshold of 0.27 rather than 0.50 (the default value in the library) was chosen in order to include more customers whose churn was confirmed. This demonstrated how the threshold can be adjusted according to the analytical objective.

The third contribution is the preparation of SHAP outputs in a structure suitable for business intelligence software. The model used 45 processed features, while the final explanations were grouped into 19 original customer variables. This diminished the technical complexity and simplified the interpretation of the explanation results in Power BI.

Further we can say that the study linked global and individual explanations within a single analytical framework. Global explanations found out the features that were most significant across the entire testing dataset whereas on the other hand the individual explanations detailed the factors linked to a specific customer prediction.

Another technical contribution is mainly the design of a six page Power BI dashboard that integrated customer profiles, churn probabilities, risk categories, actual and predicted outcomes, SHAP explanations, customer segment analysis, and model performance measures.

The dashboard structure supports different levels of analysis. Users could begin with a general overview, compare customer groups, identify high risk customers, and then examine an individual customer in much more bigger detail.

The research also contributed through the evaluation of the interactive dashboard in opposition of the static explanation baseline. The comparison considered task accuracy, completion time, questionnaire ratings, system preference, and participant feedback.

The evaluation showed how interaction could influence the way users understood and applied machine learning information. The interactive dashboard supported higher task accuracy and received stronger ratings across the main evaluation measures for the participants included in the study.

The contribution of this research is therefore not a new machine learning algorithm or explanation method. Its main contribution is the practical integration and evaluation of existing technologies within one explainable visual analytics system.

This integration demonstrates how machine learning and SHAP outputs can be transformed into information that is more accessible for users who may not have programming or machine learning experience.

Overall, the study provides a practical design example for combining predictive modelling, explainable artificial intelligence, and business intelligence visualisation. The same general approach could be adapted to other analytical problems in which users need to understand and explore model predictions.

6.4 Practical Implications

The created system has many practical applications for the companies that leverage the customer data in their decision-making processes.

The likelihood of customer churn can assist analysts in which customers may require retention attention. Instead of treating all churn cases in the same way, one can create a hierarchy of clientele depending on their risk of leaving.

The threshold value of 0.27 illustrates the flexibility of the model. Organisations for which missed churn customers are especially costly may select a lower threshold. Organisations with limited resources for retention actions may use a higher threshold and review fewer customers. The SHAP explanations provide more detailed information about contributing factors that led to the predicted risk – be it short

service tenure, month-to-month contracts, high monthly fees, or lack of any customer support.

The Power BI dashboard makes the outcomes manageable for the users who are not having the expertise in machine learning. Business analysts can apply filtering, selecting customers, comparing segments, as well as applying explanation, thus making their work more efficient.

However, the system should support human decision making rather than replace it. Predictions should be reviewed together with customer history, business costs, available resources, and other organisational information before any retention action is taken.

Overall, the system demonstrates how predictive modelling and explainable artificial intelligence can be converted into practical analytical information for customer retention planning.

6.5 Future Work

There are many ways in which we can give more to this thesis work.

In particular, the system can be evaluated with customer data from other organisations and industries to check the generalisability of the model and dashboard design beyond the selected telecommunications data set.

Other customer variables can also be incorporated such as the number of complaints, service quality, customer satisfaction, contact history, competitor offers and previous retention activity. Their impact on prediction performance and the quality of the explanations can then be assessed.

Further comparisons between more machine learning models and explanation methods can be made. The classification threshold can also be chosen using actual costs incurred by the business such as the cost of losing a customer and the cost of unnecessary retention activity.

The Power BI dashboard could be connected to a live database and customer relationship management system to evaluate automatic data updates, real time scoring, user access control and cloud deployment.

The user evaluation can be repeated on a larger, more representative participant sample. Future work should also involve professional analysts, employees working in customer retention and managers in real organisations. Statistical significance tests can

be used to investigate whether the observed differences between the static and interactive systems are statistically reliable. Future work can also investigate longer term outcomes such as learning effects, continued use, trust, decision quality and financial outcomes.

In summary, future research should investigate the use of the system with real organisational data, extend and refine the technical scope and assess the long term business impact.

6.6 Final Conclusion

In this thesis work I designed and evaluated an interactive visual analytics system for explainable customer churn prediction. The system combined machine learning, SHAP explanations, and Power BI inside one complete analytics system. The final model gave output as useful churn risk predictions, while the selected threshold gave a rise in the number of actual churn customers identified.

SHAP properly showed and explained the predictions at both global and individual customer levels. Power BI changed the predictions and explanations outputs into an understandable and interactive analytical environment. Users could examine customer risk, compare customer groups, investigate individual predictions, and also review model performance without directly using programming tools.

The user evaluation showed that the interactive dashboard supported higher task accuracy and also got higher ratings than the static explanation baseline. Most of the participants preferred the interactive system better, because it gave them permission to explore explanations and compare customer information.

In conclusion we can say that the study showed that predictions become more useful when they are supported by explanations and interactive business analytics tools. But however, the system should support human judgement rather than automatically determine customer retention decisions.

The study therefore successfully achieved its objective or goal of developing and evaluating an explainable visual analytics system for customer churn analysis.

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AI Usage Disclaimer

Generative AI tools mainly ChatGPT was used by me as a helping tool for language improvement, technical clarification, coding assistance, and document organisation. All the overall AI-assisted content was reviewed, verified, and revised by me. I am fully responsible for the final content, analysis, results, and conclusions of this thesis.

GitHub Repository

The complete project files, including notebooks, documentation, model outputs, SHAP analysis, and Power BI materials, are available in the GitHub repository:

<https://github.com/Reevegon/Reeve-Masters-Thesis/tree/main>